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Non-Adaptive Scheduling of Entanglement Purification for All-Photonic Quantum Repeaters

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To my grandparents, whose love and example have always supported and inspired me.
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Abstract

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Long-distance quantum networks need repeaters that distribute high-fidelity Bell pairs despite photon loss and operational noise. In all-photonic quantum repeaters built from Half-Repeater Graph State (HRGS), entanglement purification can be applied at various stages: during resource generation, after a single hop, or after the end-to-end pair is established. The copy count and circuit used may also differ at each stage. Choosing where to purify, how many copies to consume, and which circuit to apply is therefore a combinatorial scheduling problem. The source paper [1] demonstrates the potential of such scheduling through a single hand-picked example, but does not systematically search the space of possible schedules.

This thesis represents every fixed non-adaptive purification schedule as a Directed Acyclic Graph (DAG) and evaluates it with a single bottom-up pass computing fidelity, success probability, latency, rate, and generation cost. The evaluator reproduces the source paper's three canonical fidelity benchmarks to four significant figures. On top of this model, three search tiers – a structured baseline enumeration, an exact span-partition dynamic program, and a beam search with an added same-span pumping operation – construct and compare candidate schedules under a shared generation-cost budget $E_{\max}$. Every returned candidate is a structurally validated schedule evaluated by the same physical model throughout the protocol.

At the source paper's [1] reference configuration, a schedule found by search at the same generation cost as the hand-picked baseline improves the output rate by 2.5%. Allowing the search to use less generation cost yields a schedule requiring only 60% of the baseline generation cost while achieving a 52.8% higher output rate. A more important result concerns the effect of noise on schedule feasibility: once inner-qubit error (a form of noise not modelled in the source paper) is introduced, the fixed baseline can fall below the required fidelity, whereas a schedule re-selected by search at the same generation cost remains feasible. The same feasibility advantage is observed on average across a sample of independently randomized per-hop heterogeneous networks. Further experiments characterize how this advantage changes with noise level, timing effects, and chain length, including the minimum generation cost at which at least one searched schedule remains feasible as the chain grows.

These results are existence claims within the schedule families and search limits implemented here, not proofs of global optimality: the beam search and its pumping operation retain only a bounded frontier at each stage. Consequently, failure to find a schedule does not establish that no feasible schedule exists; a suitable schedule may remain outside the explored frontier.

Contents

Chapter 1

Introduction

1.1 Motivation

Realizing a Quantum Internet [2], [3], i.e. a network connecting quantum devices over long distances, would enable applications with no classical equivalent [4], [5]. Quantum Key Distribution (QKD) provides a shared secret key whose security is guaranteed unconditionally by the principles of quantum mechanics, independently of an eavesdropper's computational capabilities [6]–[8]. Distributed Quantum Computing (DQC) lets several small quantum processors, connected by such a network, act together as one larger one [9], [10]. Finally, networked quantum sensing can improve measurement precision beyond what independent classically connected sensors can achieve [4], [11], [12]. These applications rely on distributed entanglement as a shared quantum resource; in the repeater architectures considered here, this resource is represented by high-fidelity Bell pairs, also called *ebits*. Direct photon transmission cannot supply this resource over long distances because channel loss grows exponentially with length [4]. Quantum repeaters address this limitation [13]; Chapter 2 reviews how successive repeater generations do so.

Regardless of repeater generation, the delivered entanglement is noisy. Entanglement purification is one standard way to turn such noisy entanglement into a usable resource: it consumes several independently generated copies of a shared pair and returns fewer copies of higher fidelity [14], [15]. Each round trades additional resource consumption for a probabilistic fidelity gain. In a multi-hop repeater chain, purification can be placed at several stages: while resources are generated, after individual links are established, or at the end nodes. The number of copies consumed and the purification circuit may also vary from one stage to another. This thesis studies an all-photonic repeater architecture built from Repeater Graph States (RGSs) [16], in which purification can be inserted at different stages of resource generation, link establishment, and end-to-end connection rather than only after end-to-end entanglement is established. Choosing how to allocate a fixed purification budget across this chain is therefore a combinatorial scheduling problem. The central question is which schedule gives the best specified trade-off between rate, fidelity, and resource cost.

1.2 The Scheduling Gap

Two recent papers by Benchasattabuse, Hajdušek, and Van Meter establish the specific architecture and scheduling flexibility this thesis builds on. Throughout this thesis, [17] is called *the Bridging paper* and [1] is called *the Integrating paper*.Together, they are called *the source papers*.

The Bridging paper introduces the Half-RGS (HRGS) building block: half of an RGS, generated and held at a single station rather than requiring the full state to exist in one place, allowing an end node to act as its own source of entanglement. This enables the all-photonic scheme to interoperate with conventional memory-equipped repeater nodes instead of requiring every intermediate station to be purely photonic [17]. The Integrating paper then shows that entanglement purification can be inserted into this architecture at any stage, generation, link, or end node, and mixed freely between stages, provided it is executed *optimistically* [1], [18], i.e. without waiting for a classical heralding outcome between rounds. Chapter 3 formalises why this timing model is what makes purification worth inserting at all.

The Integrating paper demonstrates this flexibility with a single chosen schedule and identifies the search for a better one, under a fixed physical network and resource budget as an open problem. It does not establish which allocation of a fixed budget performs best, or whether that answer changes with the noise model or chain length. This is the scheduling gap this thesis addresses. Given a fixed network configuration, the research problem is to select from the combinatorially many schedules that differ in purification location, copy count, circuit, and order, then optimise a stated objective such as rate, fidelity, or cost. The working hypothesis is that systematic search finds schedules that improve on the hand-picked baseline at equal cost and, more importantly, retain feasibility in regimes where that baseline fails. This thesis tests that hypothesis across several network configurations beyond the source papers' reference configuration.

1.3 Research Objectives and Contributions

Addressing this gap requires a precise object to search over, since neither source paper formalises the schedule space, as well as search algorithms capable of exploring that exponentially large space. This thesis makes four concrete contributions toward that end.

1. A formal, DAG-based model of the schedule space for HRGS-based quantum repeaters (Chapter 4) and a bottom-up evaluator that reproduces the source paper's fidelity benchmarks for its raw, heralded, and optimistic end-node pumping schedules (defined in Chapter 5) to four significant figures.

2. A three-tier search framework: structured brute-force baselines, an exact span-partition dynamic program, and bounded same-span pumping and beam-search extensions (Chapter 5). Every returned schedule is structurally validated and evaluated using the same physical model. At the paper's reference configuration, the matched-cost result improves rate by 2.5%, while a budget-relaxed result reaches a rate of 6195.95 (the throughput metric $R(\Sigma)$ defines in Chapter 4) at cost 60, a 52.8% improvement while only using 60% of the paper baseline's budget (Chapter 6).

3. An empirical study of when scheduling changes feasibility rather than only quality, varying outer- and inner-qubit noise, the number of Bell-State Measurement (BSM) arms per hop, hop count, memory decoherence, and generation timing (Chapters 6–7). Its strongest result is a fixed-budget feasibility advantage: under non-ideal inner-qubit error, the paper schedule falls below the $F \geq 0.9$ threshold while a searched schedule at the same budget still clears it.

4. A synthesis of when and why scheduling matters, drawn from that characterization (Chapter 7): the advantage is largest in an intermediate regime, tight enough that placement decisions bind but not so tight that no searched schedule is feasible at all. In particular, within the searched families the minimum feasible budget over $N = 10$–28 is described by $E_{\max}^{\min} \approx 1.192\, N^{1.666}$. This is a descriptive fit, not an asymptotic law, and its crossover with the paper's linear budget formula $10N$ is beam-width-sensitive.

Every result above is an existence claim within the schedule families actually searched, not a proof of global optimality; Section 7.2 returns to this distinction.

1.4 Thesis Outline

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The remaining chapters proceed as follows. Chapter 2 surveys quantum repeater protocols, entanglement purification, and RGSs architectures, positioning this thesis relative to that state of the art. Chapter 3 introduces the physical objects which the rest of the thesis builds on: Bell pairs and the HRGS architecture, the error-vector and side-effect representations used to track a resource's quality through a schedule, and the purification circuits and optimistic execution model from the Bridging and Integrating papers. Chapter 4 turns these objects into a formal and searchable model: the network configuration, the state carried by each partially built resource, the schedule itself as a DAG of operations, the resource budget, and the optimization objectives this thesis considers. Chapter 5 describes the shared evaluator and the three search tiers built on top of that model. Chapter 6 reports the experimental setup, validates the evaluator against the source papers, and presents the search results across the tested regimes. Chapter 7 synthesizes when scheduling matters and when it does not, and states the scope within which the reported results should be read as optimal and the limitations of this work. Chapter 8 summarises the contributions and outlines directions for future work.

Chapter 2

Related Work and State of the Art

This chapter reviews the three main research areas on which this thesis builds: quantum repeater protocols, entanglement purification, and RGS-based architectures. It then places the scheduling gap identified in Section 1.2 in the context of the existing state of the art.

2.1 Quantum Repeater Protocols

Sending a photonic qubit directly through optical fiber or free space is limited by the channel's transmissivity, i.e. the probability that a photon reaches the receiver. For a channel of length L and attenuation coefficient α, this is $\eta = e^{-\alpha L}$. At the typical telecom C-band attenuation of $\alpha_{\mathrm{dB}} \approx 0.2$ dB/km [19], the transmissivity is already as low as $\eta \sim 10^{-40}$ over 2000 km. Thus, even with perfect sources and detectors, direct transmission becomes impractical over long distances, with the achievable rate decreasing exponentially with distance [20]. More generally, the *repeaterless secret-key capacity* of a lossy channel, bounded by the Pirandola–Laurenza–Ottaviani–Banchi Bound (PLOB), has the same exponential scaling in the high-loss regime [4], [21], [22].

A quantum repeater [13] divides a long channel into shorter elementary links, establishes entanglement over those links, and stitches them into an end-to-end pair through entanglement swapping [23], [24]. Repeater proposals are commonly grouped into three generations according to how they address photon loss and operational errors [4], [25]. First-generation (1G) repeaters use Heralded Entanglement Generation (HEG) and entanglement purification [14], [15] (Section 2.2), storing generated pairs in matter-qubit memories while other links are established [13], [26]. Second-generation (2G) repeaters replace purification with Quantum Error Correction (QEC) for operational errors while retaining heralding for photon loss [27], [28]. Third-generation (3G) repeaters use quantum error-correcting codes [29], [30] to protect against both loss and operational errors, avoiding heralding across every elementary link [16], [31].

The all-photonic quantum repeaters [16] studied in this thesis belong to the third generation but differ from proposals based on matter-qubit error correction. They encode the transmitted qubit in a multi-photon graph state [32]–[34]: the RGS [16]. Its tree encoding provides loss tolerance while operating on flying qubits, removing the need for quantum memories at intermediate stations. Section 2.3 places this architecture among related proposals; Chapter 3 gives the mechanics required here.

A related but architecturally distinct proposal, the measurement-based quantum repeater [35], stores small pre-prepared entangled resource states [36]–[40] at each station and couples them to incoming photons using simple BSMs, rather than constructing one graph state spanning the repeater chain [41]–[43]. Its emphasis on minimising the number of qubits held at each station provides a useful comparison for the resource costs studied in Chapter 6.

2.2 Entanglement Purification Protocols

Chapter 1 introduced the basic trade-off between copies and fidelity; this section covers the protocols implementing it. The two foundational two-qubit protocols, both using only Local Operations and Classical Communication (LOCC), are BBPSSW [14], which operates on Werner states, and DEJMPS [15], which extends the approach to general Bell-diagonal states without first depolarizing them. Both consume two copies and return one, succeeding only for appropriate local measurement outcomes. Thus, purification improves the quality of an entangled pair by consuming additional entangled resources, directly creating the resource-fidelity trade-off relevant to this thesis.

Entanglement pumping applies the same principle repeatedly: a working pair is purified against a sequence of freshly generated pairs, increasing its fidelity with each round [14], [15], [44]. In contrast to applying a fixed purification circuit once, pumping allows resources to be accumulated over multiple rounds while retaining one working pair. Both the resource/fidelity trade-off and pumping recur in the RGS-specific circuits on which this thesis' formal model is based (Section 3.3.1).

Later work treats the purification circuit itself as an optimisation problem rather than fixing it in advance. These approaches search over local Clifford operations for shorter circuits with higher success probability and output fidelity than the BBPSSW/DEJMPS family [45], particularly under finite resources and imperfect local operations. Purification has also been extended from bipartite Bell pairs to multipartite graph states [46], [47]. The RGS-specific two-to-one circuits used by the Integrating paper (Section 3.3.1) adapt these principles to the RGS architecture rather than directly implementing BBPSSW or DEJMPS.

Photon loss tolerance Purification addresses noise, not photon loss. Tree encoding [48] provides a different form of redundancy: instead of consuming multiple copies to improve fidelity, it encodes one logical qubit in a tree-shaped graph state whose branching factors determine its loss tolerance. Logical measurement outcomes can then be reconstructed from surviving branches. The approach was introduced for one-way measurement-based quantum computation [36], [38], [48] and later developed into a loss-tolerant quantum memory scheme [49], together with an analysis of its resource-efficiency threshold [50]. This tree encoding, rather than purification, allows the inner qubits of the RGS to tolerate photon loss (Section 3.1.2).

2.3 Repeater Graph State Architectures

The RGS was proposed as the resource state for the first all-photonic quantum repeater: a graph state whose outer qubits form entangling links to neighbouring stations while its inner qubits are tree-encoded [48], allowing the logical connection to survive photon loss [16]. Later work has focused on efficient RGS generation and tolerance to photon loss and operational errors. The original proposal generates the graph state by fusing small entangled photon sets using linear optics [16], [22], while deterministic emitter-based schemes build it directly from a small number of photon emitters, reducing the physical device count per station [51]–[53]. These approaches have also been demonstrated experimentally at small scale [54]. For loss and error tolerance, alternative graph topologies and adaptive measurement strategies can improve entanglement rates for a fixed loss and error budget [55]–[57]; code-concatenated variants instead address both through two specialized codes rather than a single tree code [28].

The Bridging and Integrating papers, discussed in Section 1.2 and formalised in Chapter 3, provide the starting point for this thesis. The former introduces the HRGS building block [17], [58], [59], while the latter shows that purification can be incorporated using optimistic (blind) execution [1],

[18]. This establishes that purification can be integrated, but not how it should be scheduled: where and how often to purify across a full chain remains open, as discussed in Section 2.5.

2.4 Schedule Optimisation in Quantum Networks

Deciding when and where to apply entanglement purification is a separate problem, even when purification is available throughout a repeater chain. In adaptive schemes, a repeater node selects its next operation based on fidelity or heralding outcomes observed so far rather than following a fixed plan [60]. Reinforcement learning and other data-driven controllers have also been proposed for quantum-network control. One study, for example, reports substantial offline costs in richer settings, including exponentially growing state representations, training times of up to ~ 36 hours per configuration, and hyperparameter searches needed to outperform a heuristic baseline [61].

Optimistic purification has likewise been studied in generic quantum networks, independently of any RGS-specific construction [18], showing that deferred heralding can be useful beyond the architecture studied here. Another scheduling decision is how qubits of a logical channel are multiplexed across serial or parallel physical links, which affects end-to-end rate and fidelity [62]. These results demonstrate that the ordering and placement of operations can influence network performance, even when the underlying physical operations are already fixed.

2.5 Positioning of This Work

The work surveyed here generally optimises a fixed repeater component, such as a purification circuit or graph-state construction, or evaluates a specified scheduling policy. For the HRGS architecture studied in this thesis, however, these decisions interact: the choice of purification location affects the resources consumed, the resulting fidelity, and the timing of subsequent operations. No systematic search considers where to purify, how many copies to consume, which circuit to use, and in what order across a full chain. As Section 1.2 establishes, this is the gap addressed by this thesis.

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Chapter 3

Background and Preliminaries

This chapter introduces the physical operations and mathematical quantities that Chapter 4 assembles into a searchable schedule model.

Section 3.1 reviews Bell pairs and the HRGS repeater architecture proposed in the Bridging paper, whose fixed protocol-level operations form what Chapter 4 calls the *backbone layer*. Section 3.2 introduces the two quantities used to track a resource's quality through a schedule: the error vector and the Z side-effect parity, together with the decoherence law that acts on the former while a resource is idle. Section 3.3 then covers the Integrating paper's purification circuits and the distinction between heralded and optimistic execution, which determines their timing cost.

3.1 Quantum Networks and Repeaters

3.1.1 Bell Pairs and Fidelity

A Bell pair is an entangled two-qubit state shared between two nodes. The canonical Bell state is

$$\left|\Phi^+\right\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle).$$

Its quality is measured by a single scalar, the fidelity $F = \langle\Phi^+|\,\rho\,|\Phi^+\rangle$, which quantifies how close the shared state ρ is to a perfect pair ($F_{\max} = 1$). Depolarising noise on the transmitted qubit and memory decoherence reduce F; photon loss instead lowers the probability that a pair can be established in the first place. Chapter 4 evaluates each finished schedule by its fidelity, rate, and resource cost for producing such a pair between the end nodes. Improving one of these quantities is meaningful only relative to this definition of F.

While Chapter 2 surveys why direct photonic transmission cannot distribute such pairs over long distances, we introduce here the elementary operation used to establish and extend links: a two-qubit Bell-State Measurement (BSM). Realized with passive linear optics and no ancillary photons, a BSM succeeds with probability at most 50% per attempt, conditional on the photons arriving [17], [42]. This ceiling is why the architecture below builds redundancy into every hop.

3.1.2 Half-RGS Repeater Architecture

A *graph state* associates one qubit with every vertex of a graph $\mathcal{G} = (V, E)$: each qubit starts in state $|+\rangle$ and a controlled-Z gate is applied along every edge of the graph [32]–[34], [38]. The result is a multipartite entangled resource whose measurement outcomes at one vertex can be used, with appropriate local single-qubit operations, to establish entanglement between other vertices [35], [36], [38]. This is the underlying resource of the architectures surveyed in Section 2.3, including the one this thesis builds on.

A *repeater graph state* (RGS) is one such graph state, with $2m$ inner qubits and $2m$ outer qubits, where m is the number of outer-qubit arms on each side. It is proposed as the basis for a memory-free all-photonic quantum repeater [16]. The inner qubits form a complete bipartite graph and are tree-encoded [48], allowing the logical measurements required by the protocol to be performed despite photon loss [50]. Each outer qubit is attached to one inner qubit and is consumed by a BSM with a neighbouring station. Such a network has two station types: Repeater Graph State Source (RGSS) nodes, which generate the RGS and hold it only until it is transmitted, and Advanced Bell-State Analyser (ABSA) nodes, which perform the BSMs between neighbouring stations and forward only the resulting entanglement, never generating anything themselves [16], [17].

The Bridging paper's own contribution is the HRGS: half of an RGS, consisting of an *anchor qubit* together with its associated inner and outer qubits, generated and held at a single station rather than requiring the full $2m$-qubit state to exist in one place [17]. Two HRGSs are combined into a full RGS by a controlled-phase gate followed by an XX measurement on their two anchor qubits, which realizes entanglement swapping – called *Swap* in this thesis – and is treated as a single operation throughout (Section 3.1.3). Because a HRGS only needs to be generated and briefly held, an end node can act as its own source: it generates a HRGS using its stationary anchor and emits the photonic qubits toward its neighbouring ABSA, rather than requiring a separate RGSS. This is what lets end-node memory scale additively with the number of hops instead of multiplicatively, the central resource-saving claim of the Bridging paper.

An N-hop chain between two end nodes has $N + 1$ stations numbered $0, \ldots, N$: stations 0 and N are the end nodes (Alice and Bob), and stations $1, \ldots, N - 1$ are RGSS nodes. Every RGSS generates a full RGS and distributes its two halves to the ABSA on each side; an end node generates only the one HRGS it needs and sends its photonic qubits to its single neighbouring ABSA. Each hop i, the link between stations $i - 1$ and i, has one ABSA that receives outer qubits from both sides and performs the requisite BSMs, so that entanglement is established one hop at a time until it spans the whole chain [17].

3.1.3 Backbone Operations and Spans

Because the BSM's success probability is capped at 50% (Section 3.1.1), each hop provisions k_i redundant outer-qubit arms rather than a single pair; Chapter 4 uses k_i as the per-hop arm-count parameter. At every ABSA, once one arm's BSM succeeds, that arm's inner qubits are *kept*: the station *discards* the redundancy by measuring every other arm's inner qubits in the Z basis [16], [17]. This *backbone selection* is forced by the protocol at every hop, on every trial.

A resource that has not yet crossed any hop – a freshly generated HRGS anchor or two such anchors combined by a Swap at the same station, for instance – is still *local* to its station of origin. Once a chain of BSMs and Swap operations connects stations across one or more hops, it forms a resource connecting stations a and b. Its *span* is written $\mathrm{Span}(a, b)$, or simply (a, b) where no ambiguity arises. The symbol κ denotes the current stage of a resource, either local to a single station or spanning some interval of the chain; Chapter 4 gives this notation a full formal treatment. Two properties of spans matter for everything that follows:

1. A resource's span is always a contiguous interval of stations, so that if it spans (a, b) then it also spans every intermediate station $a < i < b$. This is because every backbone operation that extends a span (a BSM or a Swap) connects resources across adjacent stations or at a shared endpoint.

2. A resource's span is independent of its history. Two resources that have the same span (a, b) may have reached that state by entirely different routes, e.g. one purified at every hop and the other not at all, since all that matters is the current extent of the span.

Every BSM, arm measurement, and Swap operation also produces a possible Pauli-Z side effect on the resulting resource, which must eventually be corrected. This side effect is what Section 3.2.2 tracks formally; the key fact needed here is that resolving it costs only a small, constant amount of classical communication per ABSA per trial, regardless of how deep the tree-encoded inner qubits are, since each arm's side effects collapse to a couple of bits before the ABSA passes them on [17].

3.2 Error Representation

3.2.1 The Error Vector

Just like the two source papers [1], [17], this thesis models outer-photon depolarising noise, tree-encoded inner-qubit measurement error, and memory decoherence (Section 3.2.3) as Pauli channels acting on an ideal Bell pair. Conditional on a successful transmission, the resulting state remains diagonal in the Bell basis at every stage. It is therefore represented by a real, normalized vector of four probabilities,

$$\mathbf{e} = (w, x, y, z)^{\top}, \qquad w + x + y + z = 1, \tag{3.1}$$

corresponding to the Bell-diagonal density operator

$$\rho(\mathbf{e}) := w\,|\phi_{II}\rangle\langle\phi_{II}| + x\,|\phi_{ZI}\rangle\langle\phi_{ZI}| + y\,|\phi_{ZZ}\rangle\langle\phi_{ZZ}| + z\,|\phi_{IZ}\rangle\langle\phi_{IZ}|. \tag{3.2}$$

The four components are the probabilities of no error, a Z error on the first qubit only, a Z error on both qubits, and a Z error on the second qubit only, respectively [1]. The first component w is exactly the fidelity of Section 3.1.1 and it is the quantity Chapter 4 reads off as the fidelity cost function $F(\Sigma; \mathcal{N})$. The same formalism describes a pre-transmission local pairing (an anchor and its own outer qubit, before either has crossed a hop) just as it does an established edge, which is what allows purification to be defined once and reused identically at every stage of the schedule (Section 3.3.2). This representation is closed under every operation the backbone and scheduling layers perform.

At the error-vector level, a BSM and a Swap operation apply the same bilinear composition rule to their two inputs:

$$\mathrm{BSM}(\mathbf{e}_1, \mathbf{e}_2) = \begin{bmatrix} w_1w_2 + x_1x_2 + y_1y_2 + z_1z_2 \\ w_1x_2 + x_1w_2 + z_1y_2 + y_1z_2 \\ w_1y_2 + y_1w_2 + x_1z_2 + z_1x_2 \\ w_1z_2 + z_1w_2 + x_1y_2 + y_1x_2 \end{bmatrix}, \tag{3.3}$$

which again yields a Bell-diagonal vector [1], [17]; purification (Section 3.3.2) and idle-time decoherence (Section 3.2.3) preserve the same property. Two independent contributions feed into $\mathbf{e}$ as a resource crosses a hop: an outer-qubit term from the depolarising rate e_d applied to the transmitted photon and an inner-qubit term from the tree-encoded measurement error rates $p_{\text{in}}^X, p_{\text{in}}^Z$, which contributes only to the independent x and z components, never to y [17]. Under otherwise-isotropic noise, this biases the accumulated error toward independent single-sided Z errors (x, z) over correlated two-sided ones (y), a bias that Section 3.3.1 shows the purification circuits can exploit.

3.2.2 Z Side-Effect Algebra

Independently of the error vector, every backbone operation (Section 3.1.3) may flip the sign of the surviving qubit's stabilizers, a Pauli-Z *side effect*. Unlike $\mathbf{e}$, a side effect is exactly known once its source measurement outcome is known; it therefore needs only a single bit, $s \in \mathbb{F}_2$ per resource, rather than a probability distribution. Side effects compose by eXclusive OR (XOR) whenever two resources are merged at a BSM, an arm measurement, or a Swap. This composition rule is exact, since a Pauli-Z commutes through every other operation in the model up to a sign that is tracked the same way [17]. The parity accumulated by a HRGS during its own generation, from the sequence of emitter operations used to build its tree-encoded inner qubits, is folded into this same bit before the HRGS is ever used, so no separate bookkeeping is needed for it. At the very end of a schedule, the total accumulated parity s^{total} at an end node is the XOR of every contributing side effect and a single conditional correction $Z^{s^{\text{total}}}$ applied at that point is enough to recover the intended state.

This parity bookkeeping is separate from the error vector: s determines the Pauli frame in which a resource is expressed, while $\mathbf{e}$ determines its noise. Since all side effects within an arm reduce to their XOR parity, each arm contributes only the constant-size classical message noted in Section 3.1.3, regardless of the depth of its tree encoding [17]. The scheduler tracks the parity for correctness, but compares candidate schedules through the error vector and the cost functions derived from it.

3.2.3 Decoherence Model

While idle, a resource's error vector relaxes toward the maximally mixed state at a rate set by the memory's coherence time, $T_2 = 1/\gamma$, the standard coherence-time convention [63]:

$$\mathbf{e}(t) = \tfrac{1}{4}\mathbf{1} + e^{-\gamma t}(\mathbf{e}(t_0) - \tfrac{1}{4}\mathbf{1}).$$

This relaxation is applied locally and independently at every node of a schedule to the exact idle interval that node experiences, rather than just once as a single correction at the end. Consequently, two resources that wait for different durations must be decohered before they are combined; replacing those waits by one averaged duration after composition would generally produce a different error vector [1], [17].

How much idle time a resource accumulates depends entirely on how a schedule is arranged, which is why the Integrating paper [1] singles out three canonical timing patterns that later serve, in Chapter 6, as a validation target for this thesis' own cost functions:

- *Raw*: a chain with no purification at all, so every resource is used as soon as it is generated.
- *Baseline*: each purification round is heralded before the next round begins. This avoids speculative purification but incurs a separate classical confirmation delay after every round.
- *Optimistic*: generation and purification proceed back-to-back, with intermediate confirmation deferred until the end of the sequence. Section 3.3.3 explains the resulting communication-time saving and its effect on rate.

3.3 Entanglement Purification

3.3.1 Purification Circuits: $\mathcal{C}_{\text{YY}}$, $\mathcal{C}_{\text{ZX}}$, and $\mathcal{C}_{\text{XZ}}$

Purification trades copies for quality, as surveyed in Section 2.2. In this architecture, a circuit acts on two resources at the same span κ (Section 3.1.3) through a probabilistic LOCC measurement. On success, it consumes the two inputs and leaves one output resource with an updated error vector; on failure, both inputs are discarded. The Integrating paper adapts this idea to the RGS-based resources used here [1]. Three such circuits are available, distinguished by the stabilizer they jointly measure across the two input pairs: $\mathcal{C}_{\mathrm{ZX}}$ detects a Z error on the second qubit of the pair, $\mathcal{C}_{\mathrm{XZ}}$ detects a Z error on the first qubit [15], and $\mathcal{C}_{\mathrm{YY}}$ detects the corresponding parity of Z errors across the four qubits involved [1]. Each circuit is heralded by comparing the two parties' local measurement outcomes; Section 3.3.3 distinguishes immediate from deferred confirmation. Because the inner-qubit noise term biases accumulated error toward independent single-sided errors (x, z) rather than correlated ones (y), purification sequences can exploit this bias by applying $\mathcal{C}_{\mathrm{YY}}$ first, before switching to $\mathcal{C}_{\mathrm{ZX}}$ or $\mathcal{C}_{\mathrm{XZ}}$ (Section 3.2.1).

3.3.2 Success Probabilities and Output States

The success probability and purified output vector for each circuit [1] are:

$$P_{\mathcal{C}_{\mathrm{ZX}}}(\mathbf{e}_1, \mathbf{e}_2) = (w_1 + x_1)(w_2 + x_2) + (z_1 + y_1)(z_2 + y_2), \tag{3.4}$$

$$P_{\mathcal{C}_{\mathrm{XZ}}}(\mathbf{e}_1, \mathbf{e}_2) = (w_1 + z_1)(w_2 + z_2) + (x_1 + y_1)(x_2 + y_2), \tag{3.5}$$

$$P_{\mathcal{C}_{\mathrm{YY}}}(\mathbf{e}_1, \mathbf{e}_2) = (w_1 + y_1)(w_2 + y_2) + (x_1 + z_1)(x_2 + z_2). \tag{3.6}$$

$$\mathrm{Pur}_{\mathcal{C}_{\mathrm{ZX}}}(\mathbf{e}_1, \mathbf{e}_2) = \frac{1}{P_{\mathcal{C}_{\mathrm{ZX}}}} \begin{bmatrix} w_1 w_2 + z_1 z_2 \\ z_1 w_2 + w_1 z_2 \\ x_1 y_2 + y_1 x_2 \\ x_1 x_2 + y_1 y_2 \end{bmatrix}, \tag{3.7}$$

$$\mathrm{Pur}_{\mathcal{C}_{\mathrm{XZ}}}(\mathbf{e}_1, \mathbf{e}_2) = \frac{1}{P_{\mathcal{C}_{\mathrm{XZ}}}} \begin{bmatrix} w_1 w_2 + x_1 x_2 \\ z_1 z_2 + y_1 y_2 \\ z_1 y_2 + y_1 z_2 \\ x_1 w_2 + w_1 x_2 \end{bmatrix}, \tag{3.8}$$

$$\mathrm{Pur}_{\mathcal{C}_{\mathrm{YY}}}(\mathbf{e}_1, \mathbf{e}_2) = \frac{1}{P_{\mathcal{C}_{\mathrm{YY}}}} \begin{bmatrix} w_1 w_2 + y_1 y_2 \\ x_1 z_2 + z_1 x_2 \\ y_1 w_2 + w_1 y_2 \\ x_1 x_2 + z_1 z_2 \end{bmatrix}. \tag{3.9}$$

3.3.3 Heralded versus Optimistic Purification

In the *heralded* model, each purification round waits for a full classical round-trip ($2L/c$, where L is the end-to-end fiber length and c is the signal propagation speed) to confirm the previous round's success before the next round begins. The *optimistic* (or blind) model [1], [18] instead runs generation and purification back-to-back without intermediate confirmation: operations are executed speculatively, and their classical outcomes are compared only in a deferred communication phase at the end. This replaces the n_{pur} round-trip heraldings that heralded execution would otherwise require with a single final heralding phase, which is the main source of the rate advantage reported in [1].

Crucially, *optimistic* is not a property of an individual purification operation. It follows from the placement of heralding steps relative to purification steps in the complete schedule. Section 4.3 states this criterion precisely in the formal DAG representation.

Chapter 4

The Formal Model

As established in Chapter 1 (Section 1.2), the Integrating paper [1] demonstrates purification schedules that mix generation, link-level, and end-node purification through a single hand-picked example, leaving the search for a better schedule as an open problem. Turning this into a problem that can actually be searched requires a precise object to search over. This chapter builds that object step by step: the given physical network configuration (Section 4.1), the state carried by each partially built Bell pair (Section 4.2), the schedule itself as a graph of operations (Section 4.3), the resources it is allowed to consume (Section 4.4), and finally the optimization problem this thesis addresses (Section 4.5). Section 4.6 closes the chapter by stating precisely which schedules are excluded from this model and why.

The building blocks introduced in Chapter 3 – generation, swapping, and backbone selection – are reused without redefinition below. Together, they form the *backbone layer*: the fixed, protocol-determined process of establishing entanglement. This chapter defines the *scheduling layer* built on top of it: where to insert purification, which circuit to use, in what order to apply operations, and how to compare the resulting schedules.

4.1 Network Configuration

A network configuration fixes the physical parameters that a schedule cannot change. Stations are numbered $0, 1, \ldots, N$ along the chain, so a hop is the link between consecutive stations $i-1$ and i: an N-hop chain has N hops and $N+1$ stations. With $i = 1, \ldots, N$ indexing the hops, the configuration is the tuple

$$\mathcal{N} = \Big(N,\ \{\ell_i\},\ \{\vec{b}^{(i)}\},\ \{k_i\},\ \{(p^X_{\text{in},i}, p^Z_{\text{in},i})\},\ \{\eta_i\},\ e_d,\ \gamma,\ c,\ \tau_{\text{emit}}\Big),$$

where, for each hop i, ℓ_i is its length; $\vec{b}^{(i)}$ is the list of tree-encoding branching factors at successive levels; k_i is its number of redundant BSM arms; $p^X_{\text{in},i}$ and $p^Z_{\text{in},i}$ are its inner-qubit measurement error rates; and η_i is its photon survival probability. The remaining parameters are shared across the chain: e_d is the outer-photon depolarizing rate, γ is the memory dephasing rate, c is the signal propagation speed in the fiber, and τ_{emit} is an optional per-photon emission time used to model branching-dependent generation latency. Conditional on photon arrival, the per-arm BSM success probability is capped at 50% [17], [42]. The factor η_i accounts separately for transmission loss, while k_i supplies redundancy against the measurement ceiling.

An *idealized network configuration* is one with identical hop parameters ($\ell_i = \ell$, $\vec{b}^{(i)} = \vec{b}$, $k_i = k$ for all i) and noiseless inner-qubit measurements ($p^X_{\text{in},i} = p^Z_{\text{in},i} = 0$ for all i); Section 6.3.2 relaxes the noiseless-inner-qubit assumption specifically and refers back to this definition.

Unless stated otherwise, experiments start from the Integrating paper configuration, written $\mathcal{N}_{\text{ref}}$. It has identically configured hops:

$$N = 10, \quad \ell_i = 2 \text{ km}, \quad \vec{b}^{(i)} = (16, 14, 1), \quad k_i = 18 \ \ \forall i, \quad e_d = 0.01, \quad \gamma = 0,$$
$$p^X_{\text{in},i} = p^Z_{\text{in},i} = 0 \ \ \forall i, \quad c = 2 \times 10^5 \text{ km/s}.$$

The remaining per-hop parameter, η_i, is computed from ℓ_i using attenuation $\alpha = 0.2$ dB/km, giving $\eta_i = 10^{-\alpha \ell_i/10} \approx 0.912$. The reference configuration leaves τ_{emit} unset, so the default evaluator adds no branching-dependent generation offset to the latency. Table A.1 in Chapter 6 restates the reference values above and lists, for each varied parameter, the section in which it is swept; Section 6.6 in particular explicitly enables and varies τ_{emit}.

4.2 Representing a Partially-Built Bell Pair

Every resource produced while a schedule is evaluated, from a local HRGS to the final end-to-end pair, is represented by a state:

$$S = (\mathbf{e},\ s,\ t,\ t_{\text{gen}},\ \kappa,\ r,\ h).$$

The vector $\mathbf{e}$ is the error vector introduced in Section 3.2.1, and s is the Z side-effect parity from Section 3.2.2. The remaining fields are introduced by the scheduling layer:

- t and t_{gen} are the current simulated time and the time at which the resource was first generated, respectively. Their difference, $\Delta t_{\text{idle}} = t - t_{\text{gen}}$, records the elapsed waiting time relevant to the decoherence model of Section 3.2.3.
- $\kappa \in \mathcal{K}$ is the *span* of the resource, where $\mathcal{K} = \{\texttt{RGSS}\} \cup \{\text{Span}(a, b) : 0 \le a < b \le N\}$ is the set of possible stages for a given network configuration (defined formally in Section 4.3): either `RGSS` for a resource that has not yet crossed a hop, or $\text{Span}(a, b)$ for an edge that already connects station a to station b.
- r counts how many purification rounds have already been applied to this branch.
- $h \in \{\textit{pending}, \textit{resolved}\}$ records whether the classical heralding signal for this resource has been received yet.

The field that matters most for what a schedule is allowed to do is κ: two states can only be combined by a purification circuit if they carry the *same* κ. Two branches with different histories, one purified twice and the other once for instance, can still be combined as long as both currently have the same span as previously stated in Section 3.1.3. This rule lets the model express schedules that mix purification strategies across different parts of the chain, including the type of schedule illustrated in the source paper's Fig. 4 [1].

4.3 Schedules as Directed Acyclic Graphs

A schedule is written as a pair $\Sigma = (T, \phi)$, where T is a rooted Directed Acyclic Graph (DAG) and ϕ is a labeling function

$$\phi : T \to \mathcal{K}, \qquad \mathcal{K} = \{\texttt{RGSS}\} \cup \{\text{Span}(a, b) : 0 \le a < b \le N\},$$

assigning to every node of T the span $\kappa \in \mathcal{K}$ of the state that node produces, using the same κ introduced in Section 4.2. Each node kind represents a distinct physical or logical action. The colours and shapes below form the legend used in the worked schedule of Figure 4.1:

- `GenNode` (light-blue ellipse): HRGS generation; these nodes are always leaves of T at $\kappa = \texttt{RGSS}$;
- `JoinNode` (orange box): the photonic BSM that joins the two RGSS-local HRGS resources on either side of one hop into a single-hop edge;
- `SwapNode` (green box): entanglement swapping, stitching two compatible spans (or two RGSS-local resources) into a wider one;
- `PurifyNode` (purple box): a two-to-one $\mathcal{C}_{\mathrm{YY}}$, $\mathcal{C}_{\mathrm{ZX}}$, or $\mathcal{C}_{\mathrm{XZ}}$ circuit (Section 3.3.1);
- `HeraldNode` (yellow diamond): resolving the classical heralding signal for a branch;
- `IdleNode` (grey dashed box): an explicit wait, used only where the schedule's timing requires one;
- `PauliCorrectNode` (red double circle): the root of T at $\kappa = \mathrm{Span}(0, N)$, representing the finished end-to-end pair after final Pauli correction.

Optimistic versus heralded schedules A schedule Σ is *fully optimistic* when no `HeraldNode` separates consecutive `PurifyNode` steps on any branch. In the optimistic schedules used here, all heralding is deferred to one `HeraldNode` directly below `PauliCorrectNode` at the root. A schedule is *fully heralded* when a `HeraldNode` is placed between consecutive purification rounds; each intermediate herald forces a classical round-trip ($2L/c$) before the next round can begin. Optimism is therefore a property of the schedule's timing structure, represented by the placement of `HeraldNode`s, rather than of an individual purification operation. The DAG representation can express mixed timing patterns, but the experiments compare fully optimistic and fully heralded schedules.

Legality Conditions A graph T is a *structurally legal* schedule when it has one `PauliCorrectNode` root, is acyclic, contains no unreachable nodes, and obeys the following stage constraints:

(i) `GenNode`s may only appear at $\kappa = \texttt{RGSS}$;

(ii) a `JoinNode` consumes two RGSS-local resources associated with one hop and produces $\mathrm{Span}(i, i+1)$;

(iii) a `PurifyNode`'s two children must share the same κ, i.e. the span-matching rule stated informally in Section 4.2;

(iv) a `SwapNode`'s two children must be either two RGSS-local resources or spans that share an endpoint. For the latter, i.e. (a, b) and (b, c) for some $a < b < c$, the two spans need not come from neighbouring hops, only meet at station b, and the resulting resource has span (a, c);

(v) `IdleNode` and `HeraldNode` are single-child pass-through operations: neither an explicit wait nor resolving a heralding signal changes which stations a resource spans, so both copy their child's κ unchanged;

(vi) the root must receive a state with $h = \textit{resolved}$ (Section 4.2) at $\kappa = \mathrm{Span}(0, N)$.

The implementation checks the root, graph structure, and stage consistency before evaluation, while the evaluator repeats the physical preconditions on the corresponding resource states.

This representation was checked against the source paper's own Fig. 4 schedule [1], which builds one Bell pair from purified individual links, a second from two purified 5-hop segments, and a third with no purification at all, then combines all three with a final purification round at the end nodes. It combines branches purified to different degrees and built up in different ways, and this is exactly the class of schedule this model can express: allowing κ to be an arbitrary span, and requiring `PurifyNode` to check only κ and not how each branch was built, is what makes that combination legal.

Section 6.2 confirms that this representation is not merely descriptive: evaluating the raw, heralded, and optimistic end-node pumping schedules (defined in Section 5.3) through the same bottom-up pass used elsewhere in this thesis reproduces the source paper's fidelity values to four significant figures.

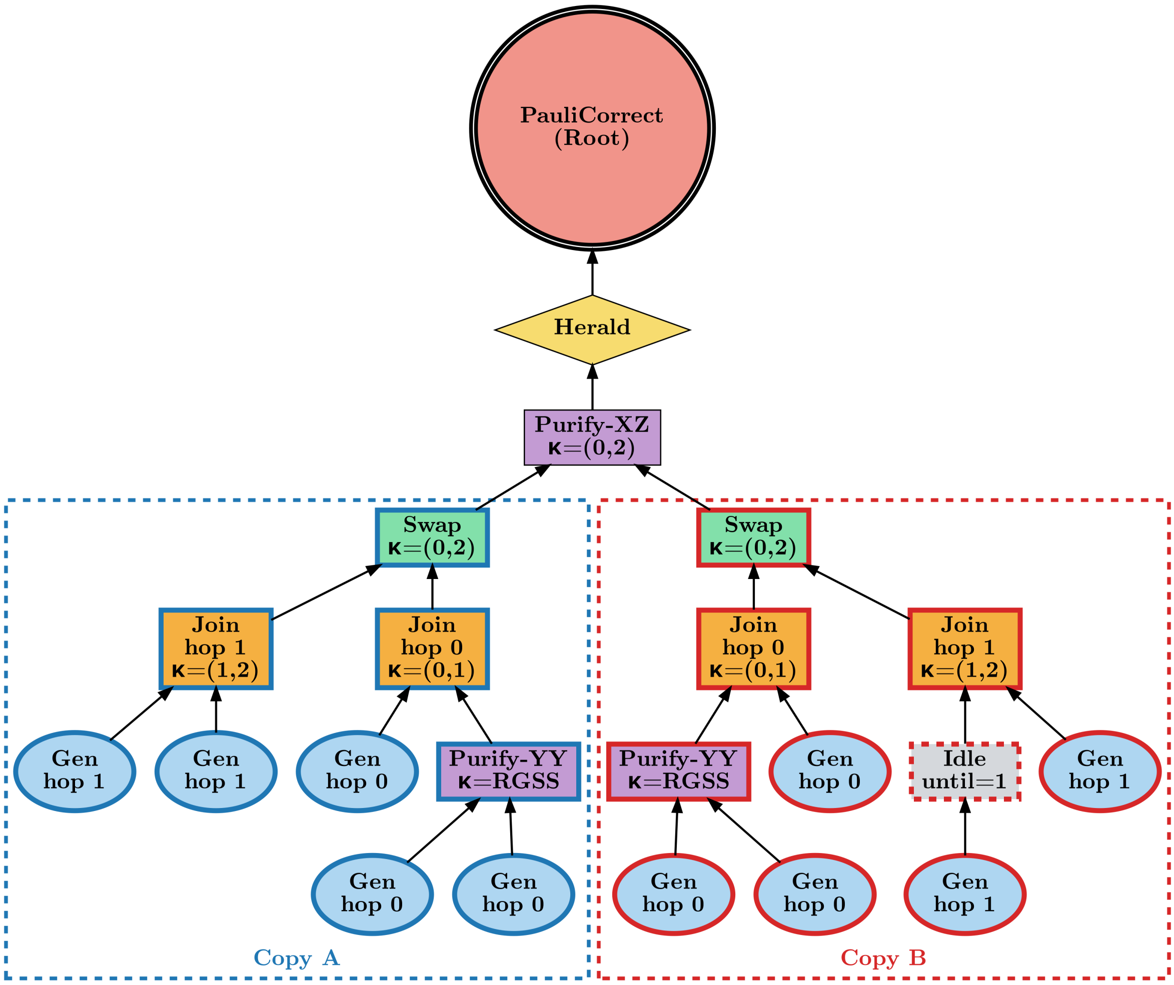


Figure 4.1: $N = 2$ example schedule Σ illustrating all seven node types ($C(\Sigma) = 10$ `GenNode` leaves). **Copy A** (left subtree): hop 0 applies `PurifyNode-YY` at $\kappa =$ RGSS before the outer-photon `JoinNode`; hop 1 is raw. **Copy B** (right subtree): identical except the right-side hop-1 `GenNode` is wrapped in an `IdleNode` (grey, dashed) modelling a synchronisation wait before the `JoinNode`. The two copies are first combined into $\kappa = (0, 2)$ by a `SwapNode`, then purified together by `PurifyNode-XZ`. The single `HeraldNode` placed after (not before) `PurifyNode-XZ` makes this an optimistic schedule (Section 4.3). `PauliCorrectNode` is the root at $\kappa = (0, 2)$ applying the final parity correction. Edges run bottom-to-top.

4.4 Resource Budget

A schedule is only meaningful together with a budget

$$B = (n_{\text{pur}},\ E_{\max},\ M_{\max}),$$

where n_{pur} specifies the number of extra HRGS copies generated for each purification round, $E_{\max}$ is the maximum number of `Gen` leaves T is allowed to contain, and $M_{\max}$ is the maximum number of branches that may be under construction at the same time.

A schedule Σ is *formally feasible* with respect to B if it uses at most $E_{\max}$ `Gen` leaves and never holds more than $M_{\max}$ branches open at once. The resource cost $C(\Sigma)$ used throughout Chapter 6 is simply the number of `Gen` leaves in T, and every headline result enforces $C(\Sigma) \leq E_{\max}$. $M_{\max}$ is enforced separately: since T only fixes a topological partial order on its operations, not a concrete real-time execution order, the number of branches simultaneously open depends on which legal execution order a scheduler chooses. For the tree-shaped candidate schedules considered here, the implementation instead computes $M(\Sigma)$ directly: the minimum peak number of concurrently open branches over every legal execution order, via the classical Sethi–Ullman register-allocation recurrence for expression trees [64]. Under this recurrence, a `Gen` leaf needs one register; `IdleNode`/`HeraldNode` inherit their child's register count because they preserve the same physical resource; and a 2-input node (`JoinNode`/`SwapNode`/`PurifyNode`) with children needing L and R registers needs $\max(L, R)$ if $L \neq R$, or $L + 1$ if $L = R$ (the classical tie case, where the first-finished side's result must be held while the second side is built). This gives the minimum register requirement for a binary expression tree, so $M(\Sigma)$ is exact for the tree-shaped schedules considered here and is computed in a single $O(|T|)$ bottom-up pass without timing information. Unlike F, R, and L, $M(\Sigma)$ depends only on the schedule structure. Σ is feasible with respect to $M_{\max}$ if and only if $M(\Sigma) \leq M_{\max}$, checked directly by the DAG-level method and, where requested, used as a constraint on the search objective (Section 4.5); Section 4.6 discusses the resulting scope limitation.

The source paper's own budget proposed convention is $E_{\max} = 10N$: five HRGS copies per side, on both sides, for every hop. This convention is used as the reference budget throughout the experiments, and Section 6.5.1 asks directly whether it remains large enough as N grows.

4.5 Objective Functions and the Optimization Problem

Four metrics can be read from the finished schedule DAG by a single bottom-up pass:

- fidelity $F(\Sigma)$, the w component of the error vector at the root (Section 3.2.1);
- rate $R(\Sigma) = P_{\text{success}}/T_{\text{attempt}}$, where T_{attempt} is the deterministic wall-clock duration of one complete schedule attempt. Under the renewal model, a failed non-adaptive purification attempt restarts the entire schedule, so the expected wall-clock time per successful pair is $T_{\text{attempt}}/P_{\text{success}}$, whose reciprocal is the stated rate;
- resource cost $C(\Sigma)$, defined in Section 4.4;
- latency $L(\Sigma)$, the makespan of T.

The optimization problem this thesis addresses is then, given a resource budget B (Section 4.4),

$$\Sigma^*(\mathcal{N}, B) = \underset{\substack{\Sigma \text{ legal w.r.t. } \mathcal{N} \\ \Sigma \text{ feasible w.r.t. } B}}{\arg\max} \; R(\Sigma; \mathcal{N}) \quad \text{s.t.} \quad F(\Sigma; \mathcal{N}) \geq F_{\min}.$$

In other words, among the legal schedules that are feasible with respect to B (Section 4.4: use at most $E_{\max}$ `Gen` leaves and never hold more than $M_{\max}$ branches open at once), find the one with the highest rate that still clears a minimum fidelity requirement. This is the objective used in every headline result in Chapter 6 unless stated otherwise, with $F_{\min} = 0.9$ matching the source paper's own threshold and $M_{\max} = \infty$ (no concurrency cap) unless a specific experiment states otherwise (Section 4.6).

The same four metrics support other research questions by swapping which one is maximized and which are constrained: maximizing F subject to a rate floor asks for the best achievable quality at a given throughput; minimizing C subject to floors on F and R asks for the cheapest schedule that still meets both requirements. Both variants were implemented as alternative objective presets while exploring these questions. Section 6.4.1 shows that switching between all three presets requires no change to the search algorithms of Chapter 5: only the scoring function and feasibility constraints need to be swapped.

4.6 Scope and Assumptions

This model rests on three simplifying choices, stated explicitly here before any algorithm or result depends on them.

First, only *non-adaptive* schedules are represented: T is fixed before execution starts and cannot branch on an intermediate measurement or heralding outcome. A fully adaptive scheme would require a different formulation, most naturally a Markov Decision Process (MDP) [65], [66]. This is left to future work (Section 8.2).

Second, $M_{\max}$ is computed exactly for any concrete Σ (Section 4.4) and can be enforced as a feasibility constraint on the search objective, but this constraint is only applied to whole, already-built candidate schedules at the point they are scored – it is not yet threaded into the recursive span-partition search of Chapter 5 as an early-pruning criterion in the way $E_{\max}$ is.

The reason is that $M(\Sigma)$, unlike $C(\Sigma)$, is not additive across independently searched sub-spans: two sub-schedules built for disjoint spans can be scheduled concurrently or sequentially, so pruning on $M_{\max}$ during search would require tracking register-count information across sibling spans, not just within each span's own subtree, risking the removal of a candidate that is only infeasible under a poor execution order. Every reported headline result therefore still optimizes without an $M_{\max}$ cap; where a cap is applied (Section 6.5.1), it changes which already-generated candidate is selected as best, not how many candidates are generated in the first place.

Third, conditional on a successful transmission, the fidelity-noise model throughout this thesis is a Pauli channel [63], [67], [68], following the assumption made by the source papers; photon loss instead reduces the success probability P_{success} rather than the fidelity F. Comparisons between schedules are therefore made under this specific fidelity-noise model, so a ranking that holds here is not automatically guaranteed to transfer to a device dominated by a different error channel. This point is returned to in Section 7.4.

Chapter 5

Search Methods

Three search tiers share one schedule representation and one evaluator. Every returned candidate passes the structural checks of Section 4.3 and is evaluated by the bottom-up pass of Section 4.5; the tiers differ only in which legal DAGs they consider, never in how a returned DAG's fidelity, success probability, latency, or cost is calculated.

The chapter moves from general search concepts (Section 5.1) through the shared evaluator (Section 5.2) to three increasingly broad but less exhaustive tiers: fixed baseline families (Section 5.3), the exact span-partition recurrence (Section 5.4), same-span pumping (Section 5.5), and beam search (Section 5.6), the heuristic used for the production-scale experiments of Chapter 6.

5.1 Preliminaries: Combinatorial Search Concepts

The schedule space defined in Section 4.3 is combinatorially large: even modest chains admit too many legal schedules to enumerate individually. The three tiers below rely on three standard tools from combinatorial optimisation, introduced briefly here before being applied to schedules.

Multi-objective optimisation and Pareto dominance A search problem is *multi-objective* when candidates are compared on several quantities at once, here cost C, fidelity F, and success probability P_{success} (Section 4.5). Candidate A *Pareto-dominates* B if A is at least as good on every quantity and strictly better on one; a *Pareto-optimal* candidate is dominated by none, and the *Pareto frontier* is the set of all such candidates [69]–[71]. Retaining the whole frontier matters here because a locally inferior candidate can become the better choice once combined with another resource, as happens when two spans are swapped together.

Dynamic programming *Dynamic programming* decomposes a large problem into smaller overlapping subproblems that are solved once and reused wherever they recur (*memoized*) [72]. Section 5.4 applies this principle to schedules: candidate sub-schedules for a short span are computed once and reused when constructing longer spans.

Heuristic and beam search When candidates become too numerous to enumerate exhaustively, a *heuristic search* trades completeness for tractability by discarding some before they are fully explored. *Beam search* does this by keeping only a fixed number of candidates, the *beam width*, at each stage, ranked by a chosen rule [73]–[75]; a narrower beam is cheaper but more likely to discard a candidate that would have led to a better final schedule. Section 5.6 applies this to the span-partition recursion of Section 5.4.

5.2 The Schedule Evaluator

The evaluator makes one bottom-up pass over a schedule DAG. At each node it applies the corresponding physical operation to the current Pauli-channel state, synchronizes branches that become ready at different times, applies the resulting memory decoherence, and propagates side effects and timing. At the root it returns fidelity F, success probability P_{success}, resource cost C, and latency L. Rate is then $R = P_{\text{success}}/L$ under the full-restart model of Section 4.5.

Section 6.2 shows that this pass reproduces the source paper's Fig. 5 fidelity numbers to four significant figures. This validates the physical model implemented by the evaluator, but does not establish that any search tier finds an optimal schedule, which is a separate question addressed by this chapter and Chapter 6.

5.3 Fixed Schedule Families

The first tier enumerates four fixed structural families:

- a *single raw end-to-end chain*, with no purification;
- *end-node pumping*, in which n_{pur} independent raw chains are purified sequentially at the end nodes with a round-trip herald between rounds (*heralded* end-node pumping);
- its *optimistic counterpart*, deferring heralding until the end (*optimistic* end-node pumping);
- *uniform link-level* purification, using the same copy count and circuit at every hop.

For even N within budget, the Integrating paper's hand-chosen schedule [1] is included as a fifth fixed comparison point.

Three of these families, the raw chain, heralded end-node pumping, and optimistic end-node pumping, are referred to throughout this thesis as the *canonical schedules*: they are the schedules whose fidelity is checked directly against the source paper's published Fig. 5 benchmarks (Section 6.2).

Within each family, purification sequences use the three two-copy circuits $\mathcal{C}_{\text{YY}}$, $\mathcal{C}_{\text{ZX}}$, and $\mathcal{C}_{\text{XZ}}$ (Section 3.3.1). All 3^r circuit sequences are enumerated up to a configured round cutoff $r_{\max}$; above that cutoff, the implementation uses a small curated set. Thus, "brute force" denotes exhaustive enumeration within these predefined structural families and circuit grid, not enumeration of every legal schedule DAG.

These fixed families serve two purposes: they provide explicit baselines throughout Chapter 6 (the raw chain and source-paper schedule as reference points, uniform link-level purification as a simple non-adaptive baseline requiring no structural search), and their candidates are always added to those produced by the two broader search tiers, so a known schedule class is never lost solely to frontier pruning.

5.4 Span Dynamic Programming

For every span $\text{Span}(a, b)$, the second tier memoizes a Pareto frontier [69] of ways to create one resource over that span. A single-hop frontier ($b = a + 1$) contains the raw link and link-level purification choices. For a wider span, the recurrence tries every split point $a < m < b$ and combines each retained candidate from $\text{Span}(a, m)$ with each from $\text{Span}(m, b)$ using a Swap operation (Section 3.1.2). Each sub-span is computed once and reused in wider constructions. This reuse, rather than repeatedly constructing and evaluating complete end-to-end schedules, is the main computational advantage of the dynamic program.

The frontier retained at each span is multi-objective over $(C, F, P_{\text{success}})$ (Pareto dominance, Section 5.1). Only non-dominated candidates are retained, following the standard Pareto-pruning rule [71]. This pruning is exact with respect to the three quantities tracked by the frontier, but does not make the search exhaustive: the caps and heuristic retention rules introduced in the next two sections can still discard a non-dominated candidate before it reaches the end-to-end span.

5.5 Same-Span Entanglement Pumping

The span-partition recurrence only builds a resource by swapping two *different* spans, so it cannot represent purifying a resource against a fresh, independently generated candidate at the *same* span κ. The second-tier extension adds this operation explicitly, bounded by configured copy-count and circuit limits rather than left unrestricted. An uncapped mode, restoring exhaustive pumping within those same limits, is available for targeted validation but can already become impractical at small N once the budget is large, so it is used only as a diagnostic, not the general production setting.

Pumping and ordinary Swap candidates compete for the same fixed-size frontier, so adding pumping can evict a swap-only candidate before it reaches an end-to-end frontier. Disabling pumping entirely reproduces the earlier pumping-free search behaviour; Section 6.3.1 reports both settings at the reference configuration since they can return different top-ranked schedules.

5.6 Beam Search

The third tier, beam search, reuses the same span recursion, node construction, and evaluator as Section 5.4, but caps every retained span frontier at a beam width w_{beam} (default 25; Appendix A.5 justifies this choice), bounding runtime and memory up to $N = 28$ hops at the cost of possibly pruning a candidate a wider beam would keep. The cap is necessary in practice: an uncapped width-7 frontier already holds roughly 1,200 candidates, making an uncapped search through the paper's $N = 10$ chain impractical. The same w_{beam} also caps the same-span pumping pool, so pumping and ordinary swap candidates share one set of slots at each span.

At each span, the w_{beam} slots split between two rankings: roughly half keep the highest-fidelity candidates, preserving purified building blocks that only pay off after a later swap, while the rest favour candidates meeting the fidelity hint $F_{\min}$, then ranked by success probability, fidelity, and cost. This split matters because a purely efficiency-based ranking would reject link-level purification at short spans, where a raw link is locally cheaper – yet accumulated noise across several such raw links can later make the end-to-end floor unattainable, which the fidelity-first half guards against.

Throughout this thesis, a *score* is the returned candidate's value of whichever quantity the active objective maximizes (Section 4.5). Under the default rate-maximizing objective used for the beam-search benchmarks of Table A.7 in Appendix A.5, score is simply the candidate's rate $R(\Sigma)$.

A different heuristic (e.g. simulated annealing or a genetic search) would need its own schedule encoding and proposal operators respecting the legality conditions of Section 4.3. Beam search instead keeps the same span-partition representation and only changes how many candidates survive at each frontier, keeping every tier on one physical evaluation path and every run *deterministic*. The result is still only a valid candidate found by a bounded search, not a guarantee of global optimality.

Chapter 6

Experiments and Results

This chapter validates the evaluator against the Integrating paper's [1] fidelity results, compares the paper schedule with schedules found by search under several network conditions, and evaluates cost-quality tradeoffs, chain-length scaling, and timing sensitivity. Additional runtime, beam-width, and supplementary sweep results are reported in Appendix A.5. Unless the Bridging paper [17] is named explicitly, "the source paper" in this chapter refers to the Integrating paper.

Throughout this chapter and the rest of this thesis, *the optimizer* refers to the software implementing one of the three search tiers of Chapter 5 that produced a given result, rather than to the schedule it returns.

6.1 Experimental Setup

Parameters and sweeps Unless stated otherwise, all experiments use the beam search with $w_{\text{beam}} = 25$ (Section 5.6) together with the fixed schedule families of Section 5.3. They start from the reference configuration $\mathcal{N}_{\text{ref}}$ (Section 4.1), notably with $N = 10$, $e_d = 0.01$, $F_{\min} = 0.9$, and $E_{\max} = 100$. Each experiment changes only the parameters specified for that experiment; all remaining parameters are held fixed. A complete summary of the experimental parameters and sweep ranges is provided in Table A.1 (Appendix A.1).

The tree branching factor is not varied because it does not affect the objectives F, C, or R introduced in Chapter 4. Uniform hop-length scaling likewise leaves relative schedule comparisons unchanged; heterogeneous hop lengths are considered separately in Section 6.3.3.

Baselines Three fixed non-adaptive baselines, introduced in Section 5.3, are used throughout: the *raw chain*, with no purification; the *uniform link-level baseline*, which applies the same copy count and circuit sequence at every hop; and the *paper schedule*, the source paper's hand-picked Fig. 4 example [1], which the paper itself presents as a demonstration of scheduling flexibility rather than an optimized design. The uniform link-level baseline and the paper schedule represent distinct fixed strategies and are labelled separately throughout the experiments.

Metrics As defined in Section 4.5, fidelity F is the Bell-pair fidelity after all purification, backbone, and correction operations, i.e. the w component of the error vector at the schedule root; rate is $R = P_{\text{success}}/T_{\text{attempt}}$, where P_{success} is the probability that a complete schedule attempt succeeds and T_{attempt} is its deterministic wall-clock duration; and resource cost C counts the number of HRGS copies, or equivalently `GenNode` leaves, consumed per trial. Two comparison modes recur throughout this chapter: *matched-cost*, in which the optimizer is constrained to $C = 100$, and *budget-relaxed*, in which it may choose any schedule satisfying $C \leq E_{\max}$.

6.2 Physics Validation

The evaluator of Section 5.2 reproduces the source paper's Fig. 5 [1] to four significant figures. At $e_d = 0.01$, the raw chain, heralded end-node pumping, and optimistic end-node pumping give fidelities consistent with the corresponding values reported by the paper. This agreement validates the physical model used by the search tiers, but does not establish search optimality. Appendix A.2 gives the corresponding comparison across the swept range of e_d.

6.3 Optimizer versus the Paper Schedule

This section first compares the search results with the source paper's schedule at the reference configuration, then examines the effect of non-ideal inner-qubit noise, outer-photon noise, and heterogeneous per-hop conditions.

6.3.1 Headline Comparison

At the reference configuration, the optimizer improves on the paper schedule both at the same cost and under the full resource budget. The matched-cost search achieves a higher rate, while the budget-relaxed search reaches a substantially higher rate at lower cost. The result obtained without the same-span pumping operation enabled is retained as a separate comparison. Detailed numerical results are reported in Table 6.1 below.

Table 6.1: Headline comparison at the reference configuration $\mathcal{N}_{\text{ref}}$ ($N = 10$, $e_d = 0.01$, $F_{\min} = 0.9$, $E_{\max} = 100$). The paper schedule is the hand-picked Fig. 4 construction from the Integrating paper [1]. Matched-cost search fixes $C = 100$; budget-relaxed search allows any schedule with $C \leq 100$. Rate is $R = P_{\text{success}}/T_{\text{attempt}}$.

Schedule	C	F	Rate	Note
Paper schedule	100	0.93	4055.92	hand-picked
Matched-cost	100	0.92	4158.14	+2.5% rate
Budget-relaxed	60	0.91	6195.95	+52.8% rate, 60% of cost
Budget-relaxed, pumping disabled	50	0.90	6713.18	lower cost still

6.3.2 Network Generalization

The optimizer was also evaluated under non-ideal inner-qubit errors and different numbers of redundant BSM arms. While the headline comparison uses the source paper's idealized network configuration, the non-ideal configurations introduce physical constraints that can change which schedules remain feasible. At the same generation budget, the paper's schedule falls below the fidelity threshold in both non-ideal configurations tested, whereas rerunning the optimizer under the corresponding physical parameters identifies feasible schedules. Figure 6.1 summarizes the resulting comparisons for the two BSM arm counts; the full numeric comparison is given in Table A.3 (Appendix A.3.1).

Rate values for an infeasible schedule are not directly comparable to those of a feasible one; Section 7.1 returns to this feasibility-versus-rate distinction as a general pattern.

6.3.3 Per-Hop Heterogeneous Networks

So far, noise has only been varied uniformly across hops (Section 6.3.2). A real deployment, by contrast, can have different link lengths, hardware quality, and error rates from hop to hop. This section assigns each hop's length, inner-qubit error rate, and arm count independently at random; the search remains non-adaptive, only the supplied network configuration changes.

A five-hop network with one hop assigned a 12× higher inner-qubit error rate than its neighbours illustrates the benefit of allocating purification resources unevenly across a heterogeneous chain, detailed in Table A.4 (Appendix A.3.2).

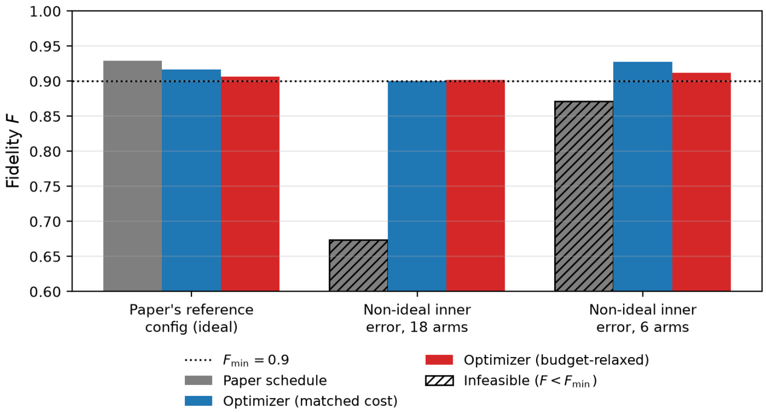
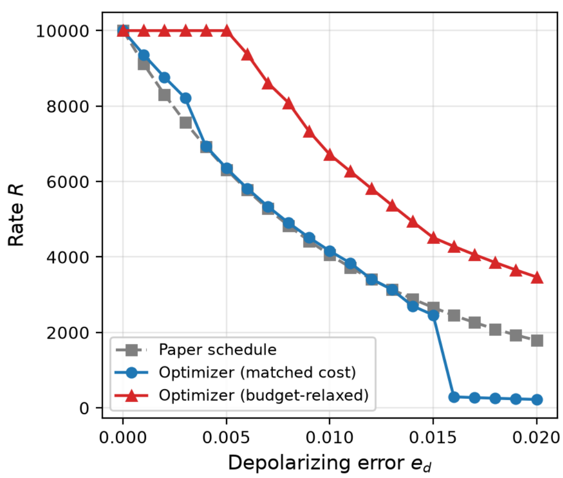


Figure 6.1: Fidelity by network configuration and schedule at $N = 10$, $e_d = 0.01$, $E_{\max} = 100$, $\gamma = 0$. Hatched bars fall below $F_{\min} = 0.9$. The paper schedule is infeasible under both non-ideal configurations; the schedules found by search remain feasible at the same budget.

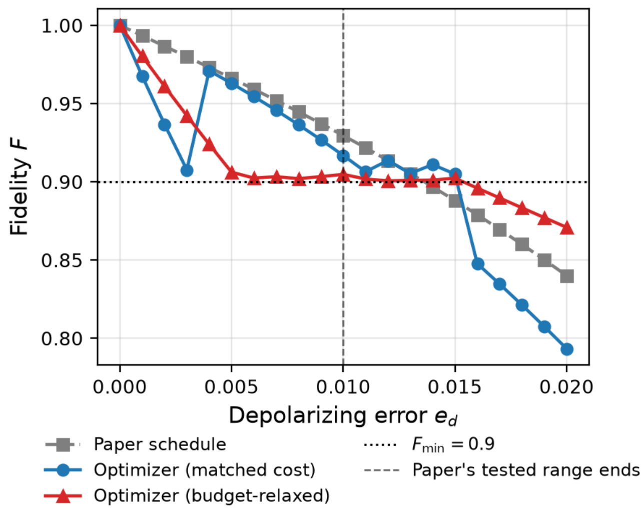
The same tendency holds over 100 independently generated random $N = 6$ networks (seeds 0–99, $e_d = 0.01$, $E_{\max} = 60$, $F_{\min} = 0.9$), reported in full in Appendix A.3.2: among the 100 networks, 80 have a feasible uniform link-level candidate; among those, the optimizer improves rate by a mean of 20.37% and never performs worse in the tested sample (Section 7.1).

6.3.4 Noise Sensitivity

Sweeping e_d from 0 to 0.01 at $N = 10$ shows that the optimizer's rate advantage over the uniform link-level baseline increases with noise. Figure 6.2 shows the rate (6.2a) and fidelity (6.2b) curves for the paper baseline, matched-cost optimizer, and budget-relaxed optimizer. At low noise the three variants remain close, with the rate curves separating visibly around $e_d = 0.004$. The matched-cost curve represents the best cost-100 result at each point, not a single fixed schedule structure. Further interpretation is given in Section 7.1.



(a) Rate vs. e_d



(b) Fidelity vs. e_d

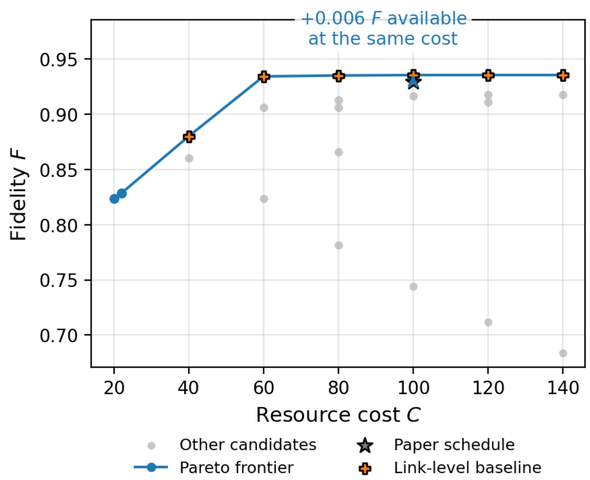
Figure 6.2: Rate and fidelity versus depolarizing error e_d at $N = 10$ and $E_{\max} = 100$ for the paper baseline, the optimizer at matched cost, and the budget-relaxed optimizer. The paper baseline's higher fidelity in the right panel reflects unused surplus above the $F_{\min} = 0.9$ floor, purchased with cost and rate the optimizer instead reallocates.

6.4 Cost-Quality Tradeoffs

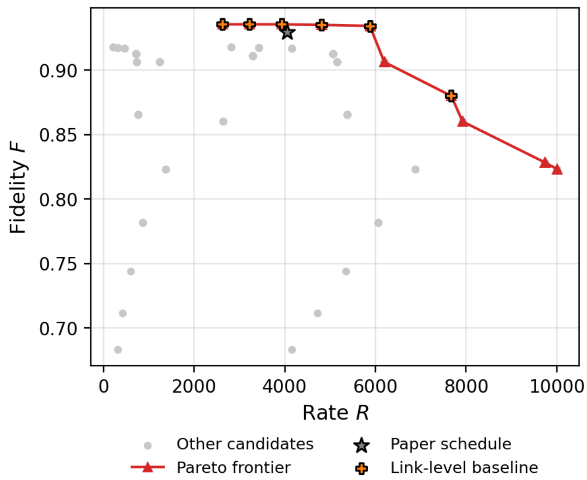
6.4.1 Pareto Frontiers

With a permissive objective ($F_{\min} = 0$, $E_{\max} = 150$), the beam search evaluates 180 candidates. Figure 6.3 shows two projections of the resulting Pareto frontier, fidelity vs. cost (6.3a) and fidelity vs. rate (6.3b), with dominated candidates and the non-dominated frontier under each pair of objectives distinguished as described in the caption (Section 5.1, Pareto frontiers).

In the fidelity-cost projection, the paper schedule lies inside the frontier, which changes little once cost exceeds $C \approx 60$. In the fidelity-rate projection, the paper schedule is again inside the frontier, which trades off high fidelity against high rate. The paper schedule and the uniform link-level baseline are marked for reference; Appendix A.7 discusses the schedule DAGs for the key optima cited here and in Section 6.3.1.



(a) Fidelity vs. cost



(b) Fidelity vs. rate

Figure 6.3: Pareto frontiers at $N = 10$, $e_d = 0.01$, $E_{\max} = 150$, over candidates evaluated under a permissive objective ($F_{\min} = 0$). Grey points are dominated; connected points form the non-dominated frontier, with the paper schedule and uniform link-level baseline marked for reference.

Alternative objectives Changing the objective from maximizing rate to maximizing fidelity or minimizing cost produces different top-ranked schedules without changing the search algorithm. Maximizing fidelity subject to a rate floor returns a higher-fidelity schedule; minimizing cost with only a fidelity floor can select a substantially lower-rate schedule. Adding an explicit rate floor changes that selection without modifying the search procedure; the complete numerical comparison is given in Appendix A.4.1.

6.5 Scaling with Chain Length

6.5.1 Budget Scaling

For N ranging from 10 to 28, the minimum $E_{\max}$ for which a schedule within the searched families reaches $F_{\min} = 0.9$ is found by bisection, rerunning the beam search at each trial budget. A descriptive power-law fit to the resulting points in log-log space gives

$$E_{\max}^{\min} \approx 1.192\, N^{1.666},$$

with an exponent greater than one over the measured range; the paper's budget is first exceeded at $N = 28$, while remaining sufficient at $N = 26$. Figure 6.4 shows the measured points together with the paper's linear budget and the fitted power law. These fitted values describe the minimum budget found within the searched families at the chosen beam width, not a global lower bound or an asymptotic result (Section 4.6); Chapter 7 discusses this scope limitation together with the excluded-construction result of Section A.4.2. Appendix A.5 additionally reports the minimum feasible budget vs. e_d at fixed $N = 10$.

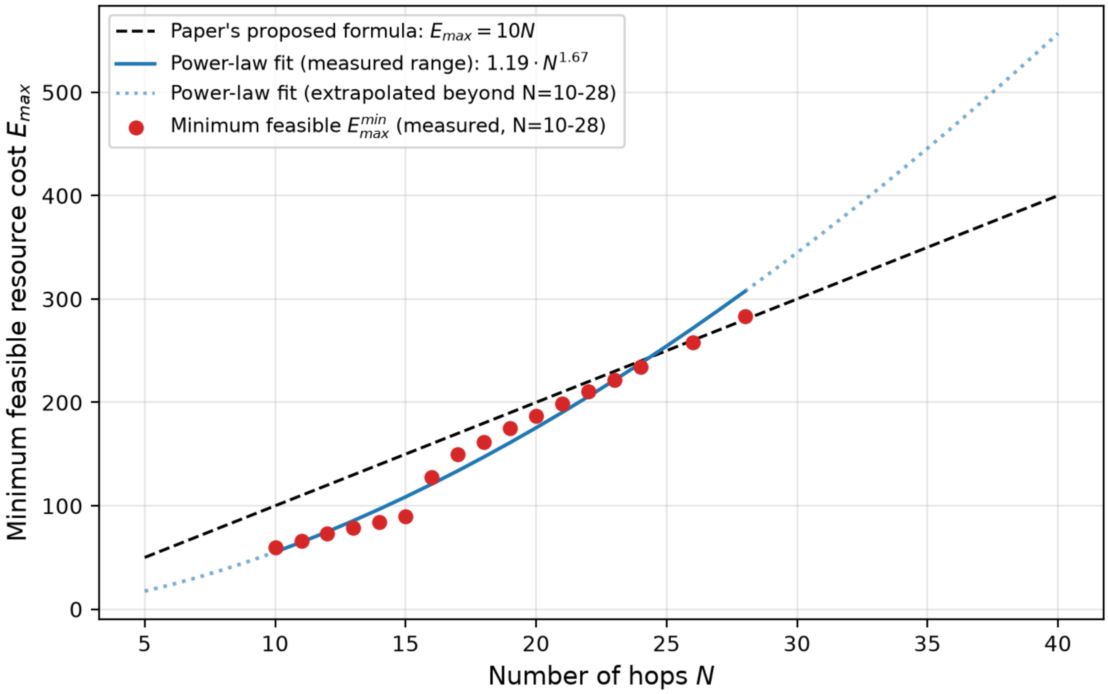


Figure 6.4: Minimum feasible $E_{\max}$ to reach $F_{\min} = 0.9$ vs. N at $e_d = 0.01$, compared with the paper's linear $E_{\max} = 10N$ budget and the fitted power law $E_{\max}^{\min} \approx 1.192\, N^{1.666}$. Solid markers and the solid fit cover the measured range ($N = 10$–28); the dotted extension is the same fit evaluated beyond that range, not additional data.

Section A.4.2 (Appendix A) additionally reports a related optimality-scope check at $N = 18$, where a hand-constructed schedule using a same-span construction excluded from the search reaches the paper's own budget.

6.6 Timing Sensitivity

Memory dephasing and generation latency affect schedules through their timing structure, including the placement of `HeraldNode`s and `IdleNode`s (Section 4.3). Sweeping the memory dephasing rate γ leaves the raw chain and paper schedule unchanged over the tested range, while heralded end-node pumping degrades as γ increases because its sacrificial copies remain in memory during the round-trip heralding delays; the schedules selected by the optimizer remain flat over the tested range because the search favours optimistic schedules. Sweeping the generation latency τ_{emit} reduces the rate of every schedule. Detailed timing curves (Figure A.7) and the corresponding parameter sweeps are given in Appendix A.6.

Chapter 7

Discussion

Chapter 6 reported results experiment by experiment. This chapter synthesizes them into a smaller number of broader claims: when and why scheduling helps (Section 7.1), the scope within which optimality claims should be read (Section 7.2), the heralded-versus-optimistic trade-off consistently exploited by the searched schedules (Section 7.3), and the limitations of the model and search procedure (Section 7.4).

7.1 When Scheduling Matters

Five results extend beyond the $N = 10$ reference configuration, subject to the search scope of Section 7.2.

Both a feasibility and a rate advantage Under the idealized noise model tested by the source papers, the paper schedule and the searched schedules are all feasible, and the gap between them is a modest rate improvement (2.5% at matched cost, Section 6.3.1). Introducing inner-qubit error, a form of noise the source papers do not model, changes this picture more substantially: the paper schedule can fall below the fidelity floor entirely, while a schedule re-selected by search at the same cost remains feasible (Section 6.3.2). In both cases the mechanism is the same: the searched schedule concentrates purification where noise has accumulated most, which a uniform hand-designed schedule cannot do by construction.

The scheduling advantage generalizes beyond hand-picked configurations A sample of 100 random per-hop heterogeneous networks confirms the same advantage beyond a single hand-chosen configuration: the searched schedule beats a uniform convention in 80 of 100 cases, by a mean of 20.37% in rate, and never does worse (Section 6.3.3). One deliberately constructed network, with a single hop assigned a 12× higher inner-qubit error rate than its neighbours, sharpens this into the same infeasible-versus-feasible pattern reported above, detailed in Table A.4 (Appendix A.3.2).

The advantage concentrates in an intermediate regime The noise sweep (Section 6.3.4) and the Pareto frontier (Section 6.4.1) both show that scheduling matters most when the fidelity constraint becomes difficult to attain. Scheduling barely matters when the fidelity floor is easy to satisfy, whether because noise is low or the allowed cost is generous: most schedules are then feasible. Their rates thus cluster together, and the searched frontier itself goes flat past a certain cost (already $C = 60$ at the reference noise level). The advantage grows as the constraint becomes more restrictive: rate improvement over the uniform baseline rises from near zero at $e_d = 0$ to 18.2% at $e_d = 0.01$, with visible separation only once e_d exceeds about 0.004. Scheduling therefore matters most when the fidelity constraint is restrictive enough to differentiate schedules, but not so restrictive that no searched schedule remains feasible.

In the noise sweep's fidelity panel (Figure 6.2b), the paper baseline sits above both optimizer variants for nearly the whole range. This is **not** a case of the paper schedule outperforming the optimizer. Every curve stays above the $F_{\min} = 0.9$ floor for most of the range, so the objective treats all three as *equally acceptable* there, and fidelity above the floor does not improve the objective, and any fidelity above the floor is *unrewarded surplus*. The paper schedule always spends the full $C = 100$ budget on a fixed circuit, whereas the budget-relaxed optimizer deliberately spends less (as little as $C = 20$ near $e_d = 0$, rising toward $C = 100$ as noise grows) and lets fidelity approach the required floor in exchange for the higher success probability and rate seen in Figure 6.2a. The two panels together show the tradeoff made by the optimizer. Extending the sweep beyond the plotted range shows that the fixed paper schedule's fidelity falls below $F_{\min} = 0.9$ by $e_d = 0.014$, while both optimizer variants remain above the floor by continually re-selecting their schedules.

The same regime reappears along chain length The minimum generation cost for $F \geq 0.9$ grows faster than the source paper's proposed linear $10N$ formula over the tested range ($E^{\min}_{\max} \approx 1.192\, N^{1.666}$, Section 6.5.1), first exceeding it at $N = 28$. This fit is descriptive, not asymptotic, and the crossover is beam-width sensitive as explained in Section 5.6. At $N = 10$, the paper's own budget already overspends by 2.0–4.8× across the tested noise range. Both observations have the same cause as the noise sweep: *a fixed fidelity floor becomes harder to sustain as the chain or the noise grows*. Section 7.2 revisits the budget-scaling result below, where the limitations of the searched families become important in interpreting the scaling.

The search accepts other objectives without requiring new algorithms None of the results above needed a different search procedure. Replacing the objective, rate, fidelity, or cost changes only which schedule is returned, not how the search finds it (Section 6.4.1), because candidate construction and evaluation are kept separate throughout Chapter 5.

7.2 Optimality Scope

The beam-search tier (Section 5.6) includes one same-span pumping operation, with the candidate frontier bounded by the beam width. All returned schedules are therefore valid constructions within the searched families, not proofs of global optimality: finding the same best candidate at two beam widths is evidence of stability, *not proof that no better candidate was pruned*. An uncapped pumping mode (Section 5.5) is available for targeted validation but becomes impractical on larger instances, so it is used as a diagnostic rather than the default.

The excluded-construction result at $N = 18$ provides the clearest demonstration. A schedule built from a purification pattern that the search never considers, by combining two independently constructed full-span candidates, is feasible within the paper's own budget (Section A.4.2). This is true even though the searched families themselves already achieve a lower minimum budget at this N (Section 6.5.1). These results do not conflict: the fact that the searched families find a feasible schedule at $N = 18$ does not imply that they can construct this particular excluded schedule, since it lies outside the search space of every tier regardless of the available budget. The same structural gap means that the budget-scaling fit of Section 6.5.1 should be interpreted as a bound on *what the searched families require*, rather than on what any legal schedule requires. This distinction is most directly relevant at $N = 28$, where the fitted bound first exceeds the paper's own formula.

The results of Chapter 6 should therefore distinguish between existence and non-existence claims. An existence claim, that a feasible schedule was found, holds for the searched model because the schedule is explicitly constructed, validated, and evaluated by the shared physical model. A non-existence claim applies only to the families and search limits actually explored; it does not prove that no legal schedule exists outside that space.

7.3 Heralded vs. Optimistic Execution

The timing results in Section 6.6 show that the relative benefit of optimistic and heralded execution depends on both memory dephasing and generation latency. The description of herald placement in Section 4.3 captures this timing trade-off. An optimistic sequence defers heralding, avoiding intermediate classical round-trips, at the cost of only identifying a failed purification after subsequent operations may already have been attempted. At the reference configuration, with no inner-qubit error and $\gamma = 0$, the optimizer *consistently favours optimistic candidates because the reduced communication delay outweighs the loss in success probability.* When $\gamma > 0$, heralded pumping is further penalized because its sacrificial copies remain in memory during each round-trip delay, while optimistic execution avoids these intermediate waits.

These two effects reinforce each other in the reference regime: the timing structure that reduces communication latency also avoids the memory waiting that would otherwise increase decoherence. This is not a universal preference: the balance depends on the relative communication delay, memory coherence, as well as failure probabilities, and different physical parameters could favour intermediate heralding instead. It should therefore be read as a property of the tested configurations, not a general architectural rule.

7.4 Limitations

1. *Non-adaptive schedules.* Only non-adaptive schedules are evaluated: the schedule is fixed before execution and cannot condition a later purification step on an intermediate outcome (Section 4.6). A fully adaptive strategy would require a different formulation, so the results bound only the non-adaptive schedule space.

2. *Memory footprint.* The search does not prune on memory footprint. $M_{\max}$, the number of branches held in memory at once, is computed exactly for every candidate (Section 4.6; quantified in Appendix A.5.1), but only the generation-cost budget $E_{\max}$ constrains the search. A schedule feasible under $E_{\max}$ may therefore still require more simultaneously held branches than a deployment with a strict memory limit could support.

3. *Noise model.* The fidelity model assumes Pauli-channel noise, consistent with the source papers. Schedule comparisons are therefore internally consistent, but their ranking need not carry over to hardware dominated by different error processes. In particular, coupled loss, gate-error, and memory effects are not represented.

4. *Fixed beam width.* The search is sensitive to which construction types compete for retained slots. Enabling same-span pumping, for example, changes which candidate is returned at the reference configuration, so adding a construction can change previously reported results even at fixed beam width. Extensions should therefore include explicit before-and-after comparisons rather than assume that a new construction can only help.

Chapter 8

Conclusion

8.1 Summary

This thesis formalized the schedule space left informal by the source papers [1], [17] and searched it systematically, addressing the four contributions set out in Section 1.3.

The Directed Acyclic Graph (DAG)-based model of Chapter 4 provides a common framework for describing and evaluating purification schedules. Its evaluator reproduces the source paper's fidelity results to four significant figures (Section 6.2), allowing different purification patterns, copy counts, and heralding placements to be evaluated under the same physical model. Built on this model, the three-tier search framework of Chapter 5 finds schedules with higher rates than the paper's hand-picked example, both at matched cost and, more substantially, when the budget is relaxed (Section 6.3.1). Since the hand-picked schedule is presented by its authors as a demonstration rather than an optimized design, this comparison primarily validates that the search produces valid, working schedules rather than constituting the thesis's strongest result.

The stronger result is instead a feasibility advantage. Under inner-qubit error, which the source papers do not model, the paper's fixed schedule can fall below $F \geq 0.9$ entirely, while a searched schedule at the same cost remains feasible (Section 6.3.2). The same pattern appears across 100 random per-hop heterogeneous networks and in a deliberately constructed weak-link case (Section 6.3.3). These results show why schedule optimization becomes important when different parts of a chain experience different conditions: a fixed construction cannot redistribute purification effort, whereas a searched schedule can.

Chapter 7 combines this pattern with the noise and chain-length sweeps to identify a common regime: the advantage is largest when the fidelity constraint is strict enough to make the choice of schedule matter, but not so strict that all searched schedules become infeasible. This regime appears across all three tested axes: noise level, chain length, and network heterogeneity. The timing experiments provide a separate observation: the preferred heralding strategy depends on the balance between communication delay, memory dephasing, and purification success probability, rather than being intrinsic to the purification operation itself.

Results Existential Claims Limitations These results are existence claims within the searched schedule families, not proofs of global optimality over the full legal schedule space (Section 7.2). The finite beam width and restricted construction families *prevent concluding that no better schedule exists*. The excluded-construction check of Appendix A illustrates this directly: it identifies a valid, feasible schedule that no search tier can construct because its construction lies outside their search spaces, regardless of how good the schedules found within those spaces are.

Taken together, the results support the working hypothesis stated in Section 1.2: systematic search can improve on the paper's hand-picked baseline at equal cost and, more importantly, retain feasibility where that baseline fails. The practical implication is that treating purification scheduling as an optimization problem, rather than fixing it by convention, can reveal useful constructions that a fixed uniform design would miss.

8.2 Future Work

The results also point to several extensions that would broaden the scope of the optimizer and reduce the gap between the current model and a deployable network.

- *Adaptive scheduling.* Conditioning later purification decisions on earlier measurement or heralding outcomes, within a single execution, could adapt to the realized state of the network rather than committing to a fixed sequence in advance. As noted in Section 4.6, this would most naturally be formulated as an MDP [65], [66], extending the current non-adaptive schedule space and potentially avoiding purification steps that become unnecessary after an intermediate outcome. Reinforcement learning is one candidate solution method, though the state and action spaces would need to be redefined carefully for this setting.

- *Online optimization.* Rather than adapting within one execution, the search itself could be re-run periodically as noise and link quality drift over minutes to months, instead of being fixed once at commissioning as it is throughout this thesis. This would allow the selected schedule to follow changes in physical conditions, particularly in heterogeneous networks where different hops may degrade at different rates.

- *Multi-path routing.* Extending the DAG formalism from a single repeater chain to tree or mesh networks would require a more general connectivity representation and a corresponding extension of the current span-based search. In such networks, purification scheduling and route selection would become coupled decisions: the optimizer could potentially choose not only where and when to purify, but also which physical path or set of paths should be used to establish the end-to-end pair.

- *Beyond the Pauli-channel model.* Hardware-specific noise models, including more detailed photon-loss and gate-error processes, would let the optimizer be evaluated against a broader range of physical implementations and could change the relative ranking of candidate schedules.

Overall, this work provides a starting point for systematic purification-schedule optimization. Even within a restricted non-adaptive space, purification placement and allocation affect both rate and the ability to meet a target fidelity. Extending the approach to adaptive decisions, time-varying networks, richer topologies, and more realistic noise models would test whether these gains persist under more realistic conditions.

Code and data availability The evaluator, the three search tiers of Chapter 5, and every experiment script behind the tables and figures in this chapter and its appendix are openly available at `https://github.com/EliottFlechtner/entanglement-purification-scheduler`.

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Appendix A

Additional Experimental Results and Supplementary Figures

This appendix collects material that supports Chapter 6 without interrupting its narrative: extended numeric tables behind figures shown in the main text, supplementary diagnostic figures for the search algorithms of Chapter 5, and one additional optimality-scope result. Sections are grouped by what they support rather than one per table, and every table or figure below is referenced from the main-chapter passage that motivates it.

A.1 Experimental Parameters

Table A.1 summarizes every parameter varied anywhere in Chapter 6, alongside the reference value used whenever it is held fixed, and the section in which it is swept. It is referenced throughout Section 6.1 and is placed first here since later sections in this appendix assume it as a common reference point.

Table A.1: Experimental parameters. The reference column gives the nominal value used unless a parameter is varied in the indicated experiment. All parameters not listed as varying follow the Integrating paper's [1] reference configuration $\mathcal{N}_{\text{ref}}$ (Section 4.1).

Parameter	Reference	Sweep range	Where varied
N (hop count)	10	10–28	Section 6.5.1
ℓ_i (hop length)	2 km	1, 2, 3, 4 km	Section 6.3.3
c (propagation speed)	2×10^5 km/s	2×10^5 km/s	Section 6.6
$\vec{b}^{(i)}$ (tree-encoding branching vector)	$(16, 14, 1)$	–	not varied
e_d (outer-photon depolarizing rate)	0.01	[0, 0.01]	Section 6.3.4
$p_X^{\text{in}}, p_Z^{\text{in}}$ (inner-qubit error rates)	0	$[0,\ 3\times10^{-3}]$	Section 6.3.2
k_i (BSM arm count)	18	6, 18	Section 6.3.2
γ (memory dephasing rate)	0	$[0,\ 10^5]$	Section 6.6
τ_{emit} (emission delay)	0	$[10^{-4},\ 10^0]$	Section 6.6
$E_{\max}$ (generation-cost budget)	100	50–280 (per experiment)	Sections 6.3.1–6.5.1

A.2 Extended Physics Validation

Section 6.2 reports the evaluator's agreement with the source paper's Fig. 5 at its single published headline point, $e_d = 0.01$. Table A.2 and Figure A.1 give the evaluator's fidelity output for the raw, heralded-pumping, and optimistic-pumping canonical schedules (Section 3.2.3) across the full $e_d \in [0, 0.01]$ grid used throughout Chapter 6, rather than only at that one point.

The source paper publishes numeric reference values at only two points on this grid, $e_d = 0.000$ and $e_d = 0.010$: approximately $\{1.000, 1.000, 1.000\}$ and $\{0.823, 0.917, 0.929\}$ respectively, for the same three schedules in the same order. The evaluator matches both to at least three significant figures, and to four at $e_d = 0.010$ (the values already quoted in Section 6.2). The nine intermediate rows have no corresponding published value to compare against, since the source paper's own Fig. 5

does not report numeric values between its two published points; they are included here as the exact evaluator output behind every noise-swept result in this thesis, not as a further reproduction check.

Table A.2: Evaluator-derived fidelity vs. e_d for the three canonical schedules of Section 3.2.3, at $N = 10$. Only $e_d = 0.000$ and $e_d = 0.010$ have a published reference value to compare against (see text).

e_d	F_{raw}	$F_{\text{baseline (heralded)}}$	$F_{\text{flexible (optimistic)}}$
0.000	1.0000	1.0000	1.0000
0.001	0.9803	0.9932	0.9933
0.002	0.9610	0.9861	0.9866
0.003	0.9422	0.9787	0.9799
0.004	0.9239	0.9710	0.9730
0.005	0.9061	0.9629	0.9661
0.006	0.8887	0.9544	0.9591
0.007	0.8717	0.9456	0.9519
0.008	0.8552	0.9364	0.9446
0.009	0.8391	0.9268	0.9372
0.010	0.8234	0.9168	0.9295

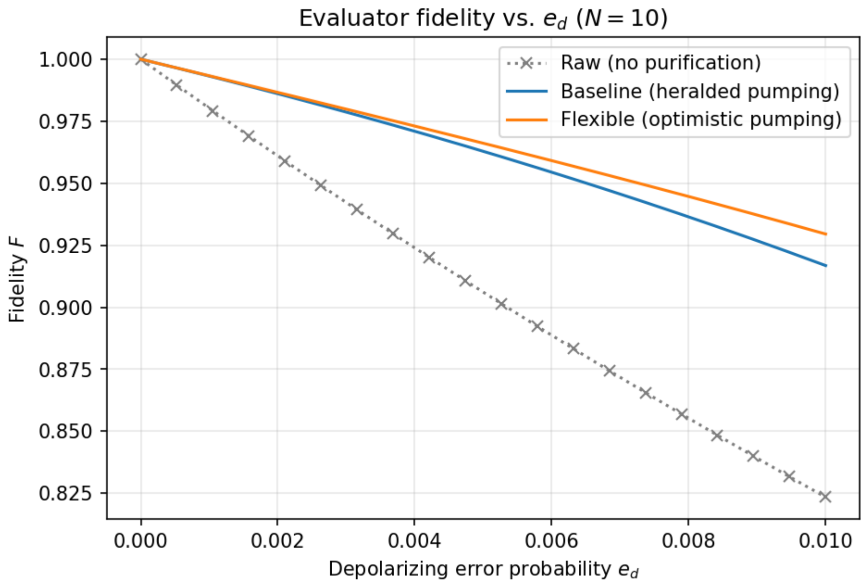


Figure A.1: Evaluator fidelity vs. e_d for the three canonical schedules of Table A.2, $N = 10$. The raw chain degrades fastest; both pumping variants track closely, with the optimistic variant consistently slightly ahead of the heralded one over the whole range (Section 7.3).

A.3 Extended Network and Heterogeneity Comparisons

This section gives the full numeric backing for the two network-generalization experiments of Section 6.3: uniformly applied non-ideal parameters (Section 6.3.2) and per-hop heterogeneous networks (Section 6.3.3).

A.3.1 Network Generalization, Full Comparison

Section 6.3.2 summarizes the network-generalization result; Table A.3 gives the full numeric comparison (cost, fidelity, and rate) behind Figure 6.1, across both tested BSM arm counts.

Table A.3: Optimizer versus the paper schedule under two non-ideal network configurations. Both use $N = 10$, $e_d = 0.01$, $E_{\max} = 100$, $\gamma = 0$, and non-zero inner-qubit error $p_X^{\text{in}} = p_Z^{\text{in}} = 3 \times 10^{-3}$, unlike the source paper's idealized inner-qubit assumption. The configurations differ in the number of redundant BSM arms per hop. A schedule is feasible when $F \geq F_{\min} = 0.9$.

Config	Schedule	C	F	Rate	Feasible
18 arms	Paper schedule	100	0.67	103.18	**no**
18 arms	Matched-cost	100	0.90	25.31	yes
18 arms	Budget-relaxed	80	0.90	85.30	yes
6 arms	Paper schedule	100	0.87	871.55	**no**
6 arms	Matched-cost	100	0.93	681.56	yes
6 arms	Budget-relaxed	60	0.91	2096.91	yes

A.3.2 Heterogeneous Weak-Link Contrast

Section 6.3.3 summarizes both the designed five-hop weak-link case and the 100-network random sample. Table A.4 gives the full numeric comparison for the designed case: one hop with a 12× higher inner-qubit error rate than its neighbours. Figure A.2 shows how the two schedules allocate resources across that chain.

Table A.4: Designed weak-link contrast for a heterogeneous $N = 5$ network at $e_d = 0.01$, $E_{\max} = 50$, and $F_{\min} = 0.9$. One hop has an inner-qubit error rate 12× higher than its neighbours. The uniform link-level baseline uses the same purification configuration at every hop, whereas the optimizer may allocate resources differently across hops. A schedule is feasible when $F \geq F_{\min}$.

Schedule	C	F	Feasible	Rate
Best uniform link-level recipe	40	0.88	**no**	1738.32
Optimizer's schedule	38	0.90	yes	1798.57

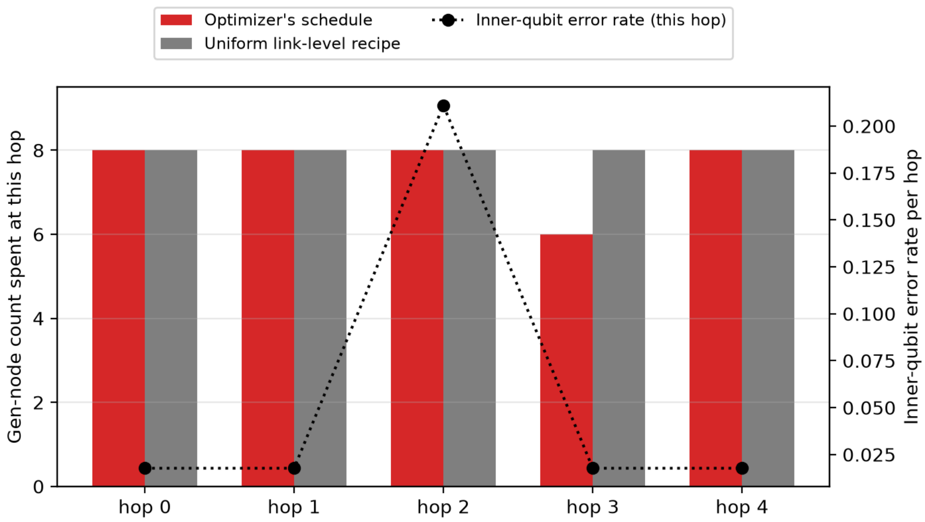


Figure A.2: Per-hop `Gen`-node allocation for the optimizer's schedule and the best uniform link-level recipe, against the per-hop inner-qubit error rate, for the weak-link network of Table A.4. The optimizer spends less at hop 3 and reallocates the saved cost to the noisier hop 2, while the uniform recipe cannot vary its allocation across hops by construction.

Figure A.3 extends this single designed case to the full 100-network random sample: the optimizer's rate improvement over the best feasible uniform recipe, plotted against each network's per-hop noise heterogeneity (coefficient of variation of inner-qubit error rate across hops). Improvement is concentrated at low-to-moderate heterogeneity, where a single noisier hop still leaves room for the optimizer to reallocate resources profitably; the points sitting on the zero line are the ties noted in Section 6.3.3, not cases where the optimizer underperforms.

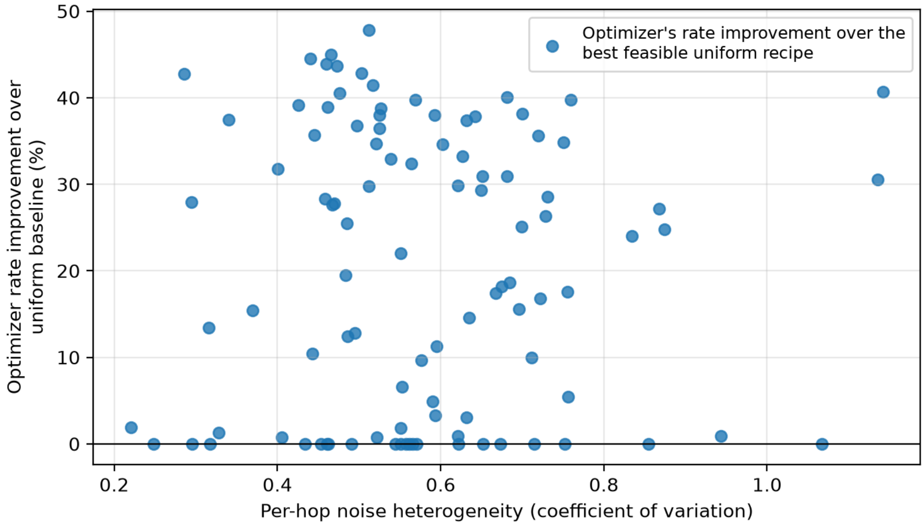


Figure A.3: Optimizer rate improvement over the best feasible uniform link-level recipe vs. per-hop noise heterogeneity, across the 100 random $N = 6$ networks of Section 6.3.3. Points on the zero line are ties, not losses (Section 7.1).

A.4 Extended Cost-Quality and Optimality Results

A.4.1 Alternative Objectives, Full Comparison

Section 6.4.1 summarizes the objective-substitution result; Table A.5 gives the full five-objective comparison it draws from, at $N = 10$ and $e_d = 0.01$.

Table A.5: Objective-substitution comparison at $N = 10$, $e_d = 0.01$: five objective definitions with no change to the search algorithm.

Objective	C	F	Rate
Maximize rate	60	0.9063	6195.95
Maximize fidelity, paper-rate floor	80	0.9351	4804.58
Maximize fidelity, matched-rate floor	80	0.9351	4804.58
Minimize cost, $F \geq 0.9$ only	60	—	1239.19
Minimize cost, $F \geq 0.9$ and $R \geq 4055.92$	60	—	6195.95

A.4.2 Optimality Bound at $N = 18$

Section 6.5.1 bisects the minimum feasible budget within the schedule families the search actually considers. This section checks that bound against one schedule outside those families, at the single point where Section 6.5.1's crossover analysis is most sensitive to it, $N = 18$.

A hand-constructed schedule using a same-span construction excluded from the search (purifying two independently constructed $\mathrm{Span}(0, 18)$ resources) achieves $F = 0.93$ at $E_{\max} = 180$, the paper's own budget for $N = 18$. The beam search cannot reach this schedule even with same-span pumping enabled (Section 5.5), because that construction lies outside the schedule families it considers – independently of whether the beam search separately finds a different feasible schedule at this N (it does, at a lower budget than $E_{\max} = 180$; Section 6.5.1). The two results are complementary, not conflicting: one shows what the searched families can already reach on their own, the other shows that a legal schedule outside those families can reach the same target through an entirely different, unexplored construction.

This is an optimality-scope result, not a scalability result: it demonstrates that the searched families do not cover the full legal schedule space, consistent with the scope stated in Section 4.6, and gives a concrete example of why the bounded-search results of Section 6.5.1 must not be read as proofs of global optimality.

A.5 Search Algorithm Diagnostics

These diagnostics justify two implementation choices used throughout Chapter 6: the default beam width $w_{\text{beam}} = 25$ (Section 5.6) and the claim that beam search scales comparably to the exact dynamic program once same-span pumping is enabled (Section 5.5).

Figure A.4 shows why $w_{\text{beam}} = 25$ was chosen: at $N = 10$, wall-clock search time grows mildly up to a beam width of about 16 and then sharply beyond it, while the best rate found is already flat across the entire tested range, so widening the beam past 25 buys no further quality improvement at this configuration. Separately, at $N = 6$, where the exact dynamic program (Section 5.4) remains tractable, beam search matches its exact optimum at every beam width tested from 1 to 32, with zero gap throughout – confirming that this determinism holds even at the smallest beam widths, not only at the default.

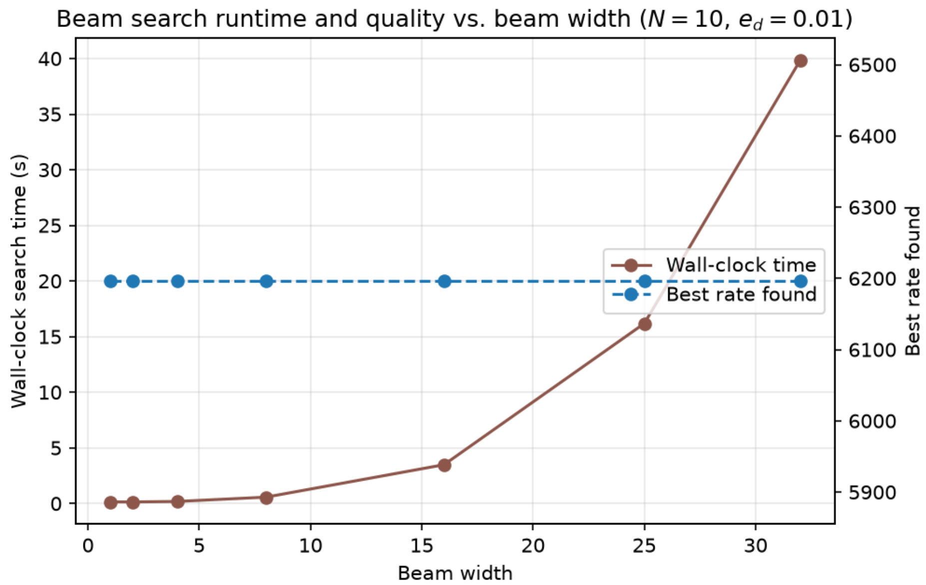


Figure A.4: Beam search wall-clock time and best rate found vs. beam width, at $N = 10$, $e_d = 0.01$.

Figure A.5 extends the runtime comparison of Table A.7 (Section A.5.2) to every tested N: both the exact dynamic program and the beam search scale similarly with N once the same-span pumping move is enabled, with the beam search consistently the faster of the two.

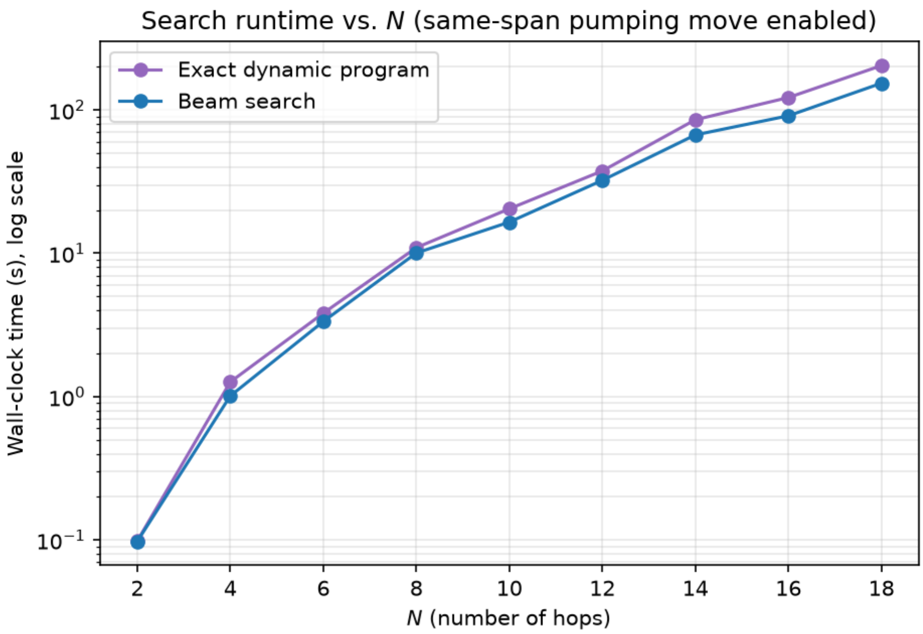


Figure A.5: Search wall-clock time vs. N for the exact dynamic program and the beam search, same-span pumping move enabled throughout.

A.5.1 Concurrent Branch Budget ($M_{\max}$)

Section 4.6 notes that $M_{\max}$, the number of concurrently open branches (Section 4.4), is computed exactly for a given schedule but is not used to prune the search itself. Table A.6 gives $M(\Sigma)$ for the fixed schedule families and the optimizer's own budget-relaxed best schedule across $N = 2$ to $N = 18$ at $e_d = 0.01$: no family shows $M(\Sigma)$ growing with N, since the register-allocation bound (Section 4.4) is governed by DAG depth and branching shape rather than hop count. Bisecting $m_{\max}$ downward from the unconstrained optimum at $N = 10$, $E_{\max} = 100$ shows that the budget-relaxed optimum ($M = 4$) remains selectable down to $m_{\max} = 4$, but no candidate in the beam-searched frontier clears both the rate objective and $m_{\max} = 3$, at which point $M_{\max}$ becomes the actual binding constraint rather than $E_{\max}$ or $F_{\min}$.

Table A.6: $M(\Sigma)$ vs. N for the fixed schedule families and the budget-relaxed optimum, $e_d = 0.01$.

N	Raw chain	Heralded end-node pumping	Optimizer (budget-relaxed)
2	3	4	3
4	3	4	4
6	3	4	4
8	3	4	4
10	3	4	4
14	3	4	4
18	3	4	5

A.5.2 Runtime Benchmarks

Table A.7 gives the wall-clock time and returned score underlying Figure A.5 and the beam-width discussion above, at representative chain lengths, together with the two cases (footnoted in the table) where the pumping-enabled exact dynamic program is not a strict upper bound on beam search (Section 5.5).

Table A.7: Wall-clock runtime and returned score for the exact dynamic program and beam search at representative chain lengths. Unless otherwise stated, each run uses the paper budget $E_{\max} = 10N$, fidelity threshold $F_{\min} = 0.9$, beam width $w_{\text{beam}} = 25$, same-span pumping enabled (Section 5.5), a 300 s per-point timeout, and a 3 GiB memory limit. The score is the value of the quantity maximized by the active objective (Section 5.6); under the default rate-maximizing objective used here, the score is the returned schedule rate $R(\Sigma)$. †At $N = 18$, the exact dynamic program completes within the timeout but finds no feasible candidate, so its score is $-\infty$, whereas the beam search returns a feasible candidate. ‡At $N = 6$, the exact dynamic program returns a lower score than the beam search. These cases occur because same-span pumping is itself beam-limited in the implementation, so the pumping-enabled exact dynamic program is not a guaranteed upper bound on the beam search (Section 5.5).

N	$E_{\max}$	Exact DP (s)	Beam search (s)	Exact DP score	Beam search score
2	20	0.10	0.10	50 000	50 000
6	60	3.82	3.34	14 372‡	15 391
10	100	20.43	16.50	6 196	6 196
14	140	85.37	66.93	3 394	3 394
18	180	203.68†	153.52	$-\infty$	1 214

A.5.3 Minimum Budget vs. Noise

Section 6.5.1 bisects the minimum feasible budget against chain length N at fixed $e_d = 0.01$. The same bisection procedure applied instead against e_d at fixed $N = 10$ (Figure A.6) gives the noise-side view behind the "2.0–4.8× overspend" figure quoted in Section 7.1: the minimum feasible budget is flat at 21 (the raw chain's own `Gen`-node floor, not a noise effect) up to $e_d \approx 0.005$, then rises to 50 by the paper's own $e_d = 0.01$, against the paper's fixed $E_{\max} = 100$.

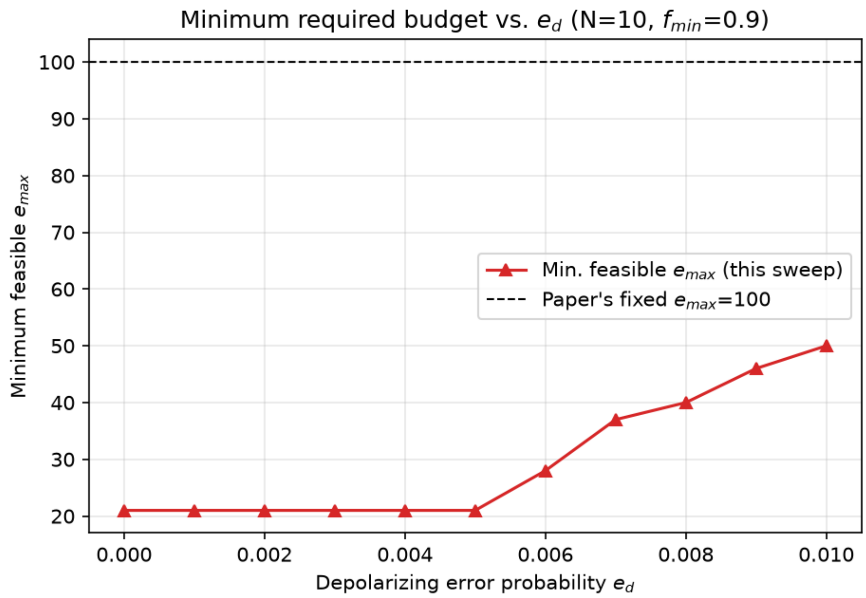


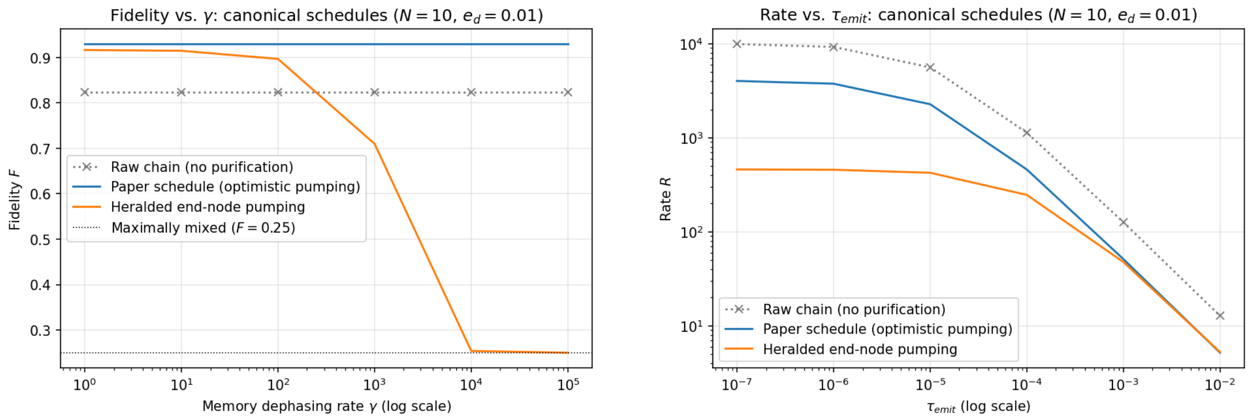
Figure A.6: Minimum feasible $E_{\max}$ to reach $F_{\min} = 0.9$ vs. e_d at $N = 10$, compared with the paper's fixed $E_{\max} = 100$.

A.6 Timing Sensitivity, Full Sweeps

Section 6.6 summarizes the effect of memory dephasing and generation latency on schedule rate and fidelity; Figure A.7 gives both sweeps in full, at $N = 10$ and $e_d = 0.01$.

Memory dephasing (γ) Sweeping γ from 0 to 10^5 leaves the raw chain and paper schedule's fidelity unchanged across the tested range: neither contains a resource that waits for another branch to become ready, so the simulated execution introduces no additional memory-decoherence interval for these schedules. By contrast, heralded end-node pumping falls from $F = 0.92$ at $\gamma = 0$ to $F = 0.25$, the maximally mixed limit, at $\gamma = 10^5$, because its sacrificial copies remain stored while each round-trip heralding delay is incurred. The schedules selected by the optimizer remain flat across the tested values of γ; this does not mean memory dephasing is generally harmless, only that the search favours optimistic schedules in which the relevant intermediate heralding delays are deferred, avoiding the specific waiting pattern that causes the collapse of the heralded pumping schedule (Section 7.3).

Generation latency (τ_{emit}) Sweeping τ_{emit} at $\gamma = 0$ reduces the rate monotonically for every schedule. The relative rate loss is largest for the optimistic pumping family, whose baseline latency is approximately $1 \times L/c$, and smallest for heralded end-node pumping, whose baseline latency is approximately $9 \times L/c$: a schedule that already has a larger fixed latency is proportionally less affected by the same additional per-photon generation delay. The two schedules cross near $\tau_{\text{emit}} \approx 0.01$; above this value, the larger fixed latency of heralded pumping makes it less sensitive, in relative terms, to further increases in generation latency.



(a) Fidelity vs. γ (b) Rate vs. τ_{emit} (log-log)

Figure A.7: Timing sensitivity of the raw chain, the paper schedule, and heralded end-node pumping, at $N = 10$ and $e_d = 0.01$.

A.7 Schedule DAGs of Selected Optima

The schedules found in Chapter 6, such as the budget-relaxed optimum at $N = 10$ (Section 6.3.1), are automatically constructed DAGs with 50 to 100 `GenNode` leaves and several hundred nodes once every backbone and purification step is counted. Rendered at a scale where individual node labels stay legible, such a diagram does not fit within a single printed page: the same rendering tooling used for the worked $N = 2$ example of Figure 4.1 produces images several thousand pixels wide once applied to these larger, automatically-constructed schedules, since node count (not page size) is the limiting factor.

Reproducing one of these diagrams here at a reduced scale would not add information beyond what Chapter 6's tables and the Pareto frontier of Figure 6.3 already convey, namely each schedule's cost, fidelity, rate, and overall shape family (e.g. end-node pumping vs. a span-partition structure), while making every individual node illegible. Figure 4.1 therefore remains the representative illustration of the DAG representation itself, at a size deliberately chosen to exercise every node type from Section 4.3 while staying legible; the full artifacts for every optimum discussed in this thesis are reproducible directly from the search calls and parameters documented in Chapter 5 and Table A.1.

A.8 Supplementary Figures for Chapter 2

[populate once figures are designed: This section is reserved for illustrative figures supporting Chapter 2 (e.g. a labeled diagram of the Repeater Graph State (RGS)/Half-RGS (HRGS) structure, or a schematic comparing first-, second-, and third-generation repeater protocols like the one in [25]). None of the related-work discussion in Chapter 2 currently depends on a figure placed here; add the figure, a short caption, and a forward pointer from the relevant paragraph of Chapter 2 to this section once the figures are designed.]